Team 106 Problem B

Bendin' It Like Bayes: Predicting Where a Penalty Kick Enters the Goal from Two Early Snapshots

Summary

From two positions of a penalty kick, at 0.05 and 0.10 s, we predict where the ball enters the goal, with its uncertainty.

We assume still air, a rigid regulation ball, drag and lift laws from wind-tunnel and trajectory measurements, and rounding as the only observation error. The flight model includes gravity, drag with its crisis, the Magnus force, spin decay and knuckling. Six coordinates cannot fix its unknowns, so plausible kicks are weighted by their fit to the observations and flown to the goal (Bayesian inference).

The ball most likely enters the goal high on the goalkeeper's right, at $y^* = (2.67 \pm 0.10)$ m, $z^* = (1.97 \pm 0.13)$ m (± 1 s.d.; total with the spin-prior systematic: ± 0.11 m in y, ± 0.13 m in z; 68 % region ± 0.15 m in y and ± 0.20 m in z, 95 % region ± 0.25 m in y and ± 0.35 m in z); it enters cleanly with probability 0.999 (0.960 allowing for model error); it crosses the goal line (0.392 ± 0.003) s after the kick, leaving the goalkeeper 0.29 s after the second observation, less than the time an instantaneous dive needs to reach it and far less than the 0.6 s measured for real dives.

In simulated tests the method's median error matches an exact spin fit's, but it never fails, and its 68 % and 95 % regions contain the truth in 68 % and 92 % of kicks. If the physics lies as far outside our priors as in the mismatch tests, the regions must widen by 1.64 (± 0.25 m in y, ± 0.33 m in z at 68 %). A goalkeeper told the answer at the second observation would save it with probability 0.018: the limit is time, not prediction.

Contents

1 Introduction **4**
1.1 Background . . . 4
1.2 Problem Restatement and Interpretation . . . 4
1.3 Overview of Our Approach . . . 5

2 Assumptions **6**
2.1 The Ball and the Kick . . . 6
2.2 Observations . . . 7
2.3 Environment . . . 7
2.4 Aerodynamics . . . 7

3 Notation **8**

4 Physical Model **8**
4.1 Equation of Motion . . . 8
4.2 Aerodynamic Forces . . . 10
4.3 A Hierarchy of Models . . . 11
4.4 What the Two Observations Already Say . . . 13
4.5 Which Effects Matter . . . 14

5 Prediction Method **14**
5.1 The Idea . . . 14
5.2 Unknowns and Priors . . . 15
5.3 Algorithm . . . 15
5.4 Alternative Predictors . . . 17
5.5 Inputs, Outputs and Domain of Validity . . . 17

6 Uncertainty Estimation **18**
6.1 Sources of Uncertainty . . . 18
6.2 From Samples to Regions . . . 18
6.3 Independent Check: the Delta Method . . . 18
6.4 Which Uncertainty Dominates . . . 19
6.5 Calibration . . . 20

7 Testing the Method **20**
7.1 Synthetic Ground Truth . . . 20
7.2 Accuracy . . . 22
7.3 Calibration . . . 22
7.4 Noisier Data and Model Mismatch . . . 22
7.5 The Configuration Used for the Given Kick . . . 24
7.6 Failure Modes . . . 24

8 Application to the Given Scenario **25**
8.1 The Predictive Density . . . 25
8.2 All Methods on This Kick . . . 25
8.3 The Kick Itself . . . 27
8.4 Can the Goalkeeper Save It? . . . 27

9 Sensitivity Analysis **27**

10 Strengths and Weaknesses **30**
10.1 Strengths 30
10.2 Weaknesses 30

11 Conclusion **31**

A Details of the Flight Model **33**
A.1 Numerical Integration 33
A.2 Calibration of the Knuckling Force 34

B Inference Details **34**
B.1 The Spin Prior in Terms of the Lift Vector 34
B.2 Adaptive Proposal for the Nuisance Parameters 35
B.3 Deterministic-Mixture Weights 35
B.4 Knuckling in the Observation Window 35
B.5 Further Test Results 35

Report on the Use of AI **37**

1 Introduction

1.1 Background

A penalty kick is decided faster than a goalkeeper can move. Penalty kicks leave the foot at about $26\,\mathrm{m\,s^{-1}}$ to $32\,\mathrm{m\,s^{-1}}$ on average (Morya et al., 2005) and reach the goal line in well under half a second, while a dive to the side takes at least 0.6 s (van der Kamp et al., 2018). Whatever can be known about the ball's destination has to be read from the kicker or from the first tenths of a second of flight.

The flight is not a parabola. At these speeds a football has a Reynolds number of order 10^5, the regime of the *drag crisis*, in which the drag coefficient falls from about 0.5 to about 0.2 or less (Hong and Asai, 2014). A spinning ball feels the Magnus force, perpendicular to its velocity; a ball struck with almost no spin *knuckles*, pushed sideways at random: in wind tunnels its side and lift forces fluctuate with standard deviations of up to about 3 N (Hong and Asai, 2014). Over the 11 m to the goal, gravity and the Magnus force each move the ball by the order of a metre (Section 4.5).

The problem gives two positions of the ball, at 0.05 s and 0.10 s. With the known starting point these are six numbers, enough to fix a launch velocity and one constant acceleration, and nothing more. A faithful model of the flight has many more unknowns: the spin vector, the drag and lift curves of this particular ball, the air density and the ball's mass. The data cannot determine these, but wind-tunnel and trajectory experiments constrain them. Predicting the entry point is therefore an inference problem: six numbers must be combined with prior physical knowledge into a probability distribution for where the ball crosses the goal plane. We build that combination explicitly, test it on simulated kicks whose true destination is known, and apply it to the given kick.

1.2 Problem Restatement and Interpretation

We use the coordinates of the problem statement (University Physics Competition, 2026): origin at the penalty mark, x from the mark to the goal, z up and y completing a right-handed system, so $+y$ is the kicker's left and the goalkeeper's right. The goal is the rectangle $|y| < W/2$, $0 < z < H$ in the plane $x = L$, with $L = 11\,\mathrm{m}$, $W = 7.32\,\mathrm{m}$ and $H = 2.44\,\mathrm{m}$. The kick takes place at $t = 0$ with the centre of the ball at $\mathbf{r}_0 = (0, 0, 0.11)\,\mathrm{m}$; we are given $\mathbf{r}_1 = \mathbf{r}(t_1)$ and $\mathbf{r}_2 = \mathbf{r}(t_2)$ at $t_1 = 0.05\,\mathrm{s}$ and $t_2 = 0.10\,\mathrm{s}$. The problem asks for

(T1) a general method that, given $\mathbf{r}_1$ and $\mathbf{r}_2$ for any kick, predicts the point at which the ball is most likely to enter the goal;

(T2) an estimate of the uncertainty of that prediction;

(T3) tests showing how accurate the prediction is and whether its stated uncertainty can be trusted;

(T4) the prediction and its uncertainty for the given kick, $\mathbf{r}_1 = (1.50, 0.45, 0.44)\,\mathrm{m}$ and $\mathbf{r}_2 = (2.96, 0.86, 0.74)\,\mathrm{m}$.

Several terms need an operational definition.

Entry point.
The point (y^*, z^*) where the centre of the ball crosses the plane $x = L$, at time t^*. The ball enters cleanly if its centre satisfies $|y^*| < W/2 - r_b$ and $r_b < z^* < H - r_b$, with r_b the ball's radius; a centre within r_b of the frame strikes a post or the bar.

Most likely.
The maximum (mode) of the probability density $p(y^*, z^* \mid \mathbf{r}_1, \mathbf{r}_2)$ of the entry point given the observations, the *predictive density*. The problem asks for the mode, not the mean: if the density is skewed, the mean can lie where the ball is unlikely to go.

Uncertainty.
The 68 % and 95 % highest-density regions (the smallest regions that contain that probability) and the standard deviations of y^* and z^*. An uncertainty is useful only if it is honest, so we test *calibration*: on simulated kicks, the 68 % region should contain the true entry point 68 % of the time.

General.
The method accepts any $(\mathbf{r}_1, \mathbf{r}_2)$ and contains nothing tuned to the given kick; its domain is set by Table 2: launch speeds of $15\,\mathrm{m\,s^{-1}}$ to $40\,\mathrm{m\,s^{-1}}$ and balls that meet the Laws of the Game.

Observations.
The positions are given to 0.01 m; we read them as true positions rounded to that precision (Assumption A4) and test larger errors separately.

We predict where the ball would cross the goal plane if nobody touched it, and compare the flight time with the time a goalkeeper needs to dive, because that decides whether the prediction can be acted on.

1.3 Overview of Our Approach

For the given kick the ball goes high on the goalkeeper's right, inside the frame (Section 8), and the prediction comes too late to act on: the ball crosses the goal line 0.29 s after the second observation. Figure 1 shows the geometry, the two observations and the predicted flight. The method has three layers.

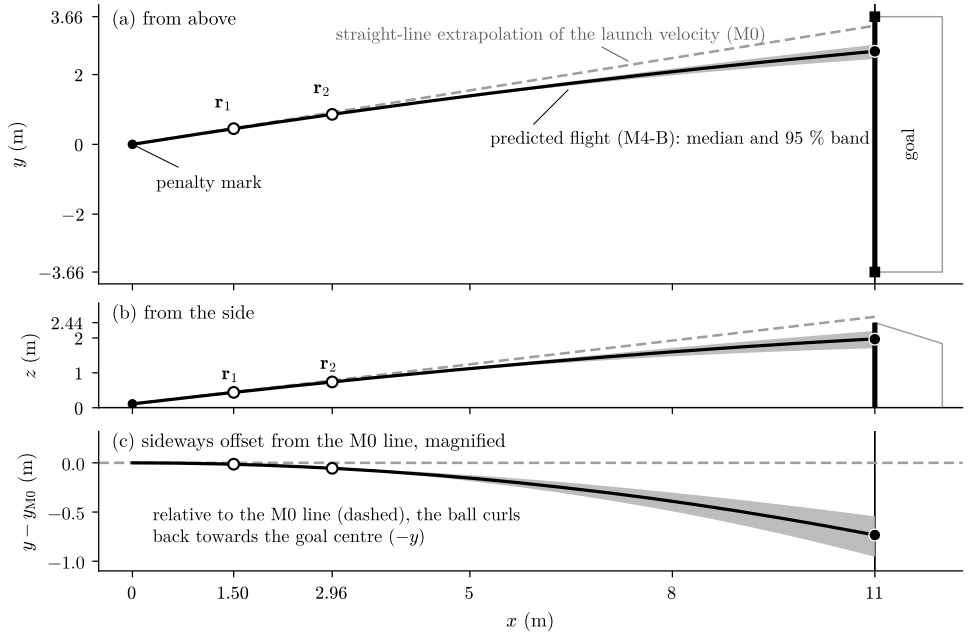


Figure 1: The two observations cover the first 3 m of the 11 m flight, seen from above (a) and from the side (b), where y and z are drawn at half the scale of x. The predicted flight (black: median of the posterior trajectories; grey: 95 % band) bends back towards the centre of the goal and drops under the bar, while the straight-line extrapolation of the launch velocity (M0, dashed) clears the bar. (c) Sideways offset of the predicted flight from the M0 line, magnified about three times: the curl back towards the centre grows from centimetres at the observations to most of a metre at the goal.

1. **Physics** (Section 4). A hierarchy of flight models runs from the straight-line extrapolation of the launch velocity (M0) to a full model (M4) with gravity, drag that depends on Reynolds number and spin, a spin-dependent Magnus force, spin decay and a random knuckling force calibrated to wind-tunnel measurements. The observations alone already show spin: the ball accelerates sideways at $(11.1 \pm 2.5)\,\mathrm{m\,s^{-2}}$, more than four standard errors and far more than knuckling can explain. Its drag they fix only together with the exact instant of the kick.

2. **Inference** (Sections 5 and 6). We sample many possible kicks (launch velocity and instant, spin, drag and lift curves, air density, ball mass), weight each by how well it reproduces the two observations, and fly it to the goal plane. The peak of the weighted cloud of crossing points is the prediction, its spread the uncertainty; the delta method, which uses no prior, checks the spread.

3. **Testing** (Section 7). On simulated kicks with known entry points, we compare the method with ten alternative predictors, from M0 to an exact fit of a constant-coefficient spin model and four predictors learned from simulated kicks. In the configuration used for the given kick, with the kick instant exact (Section 7.5), its median error, 0.18 m, matches the exact fit's and is far below M0's (0.87 m); unlike the fit it never fails, and its 68 % regions contain the true entry point in 68 % of the kicks.

Section 8 applies the method to the given kick, Section 9 varies its assumptions, and Section 10 discusses its strengths and weaknesses.

2 Assumptions

Each assumption is numbered and justified; where one neglects an effect, we give the size of what is neglected (Table 5). Assumptions that carry the result are varied in Section 9.

2.1 The Ball and the Kick

A1 Rigid regulation ball. The ball is a rigid sphere of radius $r_b = 0.11\,\mathrm{m}$ and mass m between 0.41 and 0.45 kg, with its centre of mass at its centre.
Justification. The problem places the centre of the ball 0.11 m above the ground; the implied circumference $2\pi r_b = 0.691\,\mathrm{m}$ lies inside the 0.68 m to 0.70 m of Law 2, which also sets the mass range (The International Football Association Board, 2026). By the first observation the ball is 1.5 m from the mark, long after contact with the foot, so it no longer deforms.

A2 Impulsive launch at the labelled instant. The ball is in free flight from the labelled kick instant on, $\mathbf{r}(\tau) = \mathbf{r}_0$, with $\tau = 0$ for the given kick. For tracking data whose kick is labelled only to a camera frame, the method accepts a two-sided offset τ, Gaussian with standard deviation $\sigma_\tau = 2\,\mathrm{ms}$.
Justification. The problem states the kick instant exactly. A frame-labelled kick can be off by milliseconds either way; the width is our assumption. The offset trades against the drag and does not change the answer (Section 4.4).

A3 Unobstructed flight. Between launch and the goal plane the ball touches nothing: not the ground, the goalkeeper or the frame.
Justification. A goalkeeper needs to know where the ball would go if he did nothing. A simulated centre that falls below $z = r_b$ before $x = L$ counts as ground contact.

2.2 Observations

A4 Rounding is the only observation error. The given coordinates are the true positions of the ball's centre rounded to 0.01 m. The rounding errors are independent and uniform on ± 5 mm, with standard deviation $\sigma = 0.01\,\mathrm{m}/\sqrt{12} = 2.9\,\mathrm{mm}$. In the likelihood (Section 5) we replace the uniform distribution by a Gaussian of the same standard deviation. The starting point $\mathbf{r}_0$ is exact.
Justification. The problem states no measurement error, so the stated precision is the only information about it. Section 7 tests the Gaussian approximation on kicks whose only error is rounding (median error 0.19 m, 68 % coverage 63 %) and repeats the tests with $\sigma = 5$, 10 and 20 mm. This assumption carries the size of the uncertainty: at those errors the standard deviation of the given kick's y^* grows from 0.10 m to 0.17 m, 0.24 m and 0.33 m (Section 9).

2.3 Environment

A5 Uniform gravity, non-rotating frame. Gravity is $g = 9.81\,\mathrm{m\,s^{-2}}$, vertically down, everywhere on the pitch.
Justification. The Coriolis acceleration $2\Omega v$, with $\Omega = 7.29 \times 10^{-5}\,\mathrm{rad\,s^{-1}}$ the rotation rate of the Earth, is $0.005\,\mathrm{m\,s^{-2}}$ and moves the ball by about 0.4 mm before it reaches the goal, less than the 2.9 mm precision of the data.

A6 No buoyancy.
Justification. The buoyant acceleration $\rho V g/m$, with V the ball's volume, is $0.15\,\mathrm{m\,s^{-2}}$, or 1.6 % of g. It raises the entry point by about 13 mm, an order of magnitude below the uncertainty of the prediction.

A7 Still air. The predictor assumes no wind.
Justification. The problem gives no wind, and two observations 0.05 s apart cannot distinguish a steady crosswind from a slightly different launch direction. A $2\,\mathrm{m\,s^{-1}}$ crosswind pushes the ball sideways at about $1.3\,\mathrm{m\,s^{-2}}$ and moves a trajectory with a given launch by about 0.10 m; but the observations already contain that push, so re-inferring the given kick with the wind known moves the answer by only 0.036 m, not resolved from the Monte Carlo noise (Section 9). Across kicks an unknown wind still costs accuracy, so it is part of the mismatch test (Section 7).

A8 Incompressible air of uncertain density. The air density ρ lies between 1.05 and $1.25\,\mathrm{kg\,m^{-3}}$; the kinematic viscosity is $\nu = 1.54 \times 10^{-5}\,\mathrm{m^2\,s^{-1}}$ (Goff and Carré, 2012).
Justification. The range of ρ is our assumption and covers stadiums from sea level to about 1500 m altitude. The Mach number is $v/c \approx 0.09$, where $c = 343\,\mathrm{m\,s^{-1}}$ is the speed of sound, so compressibility is negligible.

2.4 Aerodynamics

A9 Quasi-steady forces. At each instant, drag and lift are set by coefficients $C_D(\mathrm{Re}, \mathrm{Sp})$ and $C_L(\mathrm{Sp})$ of the instantaneous Reynolds number and spin parameter, as measured in wind tunnels and trajectory experiments (Hong and Asai, 2014; Goff and Carré, 2009, 2012).
Justification. Air passes the ball in about 7.0 ms ($2r_b/v$), much less than the constant-acceleration flight time of about 0.41 s, so the flow adapts to the slowly changing speed and spin. The one unsteady effect known to matter, knuckling, is added separately (Assumption A12).

A10 Unknown ball design. The drag and lift curves of the ball being kicked are not known. We describe them by the parameter ranges of Table 2, which span published values for modern match balls, and average the prediction over these ranges.
Justification. Hong and Asai (2014) found that the shape and orientation of the panels, more than their number, move the drag crisis between Reynolds numbers of about 2.2×10^5 and 3.6×10^5; we extend the range down to 1.5×10^5, our assumption, to cover older and smoother designs.

A11 Slow spin decay about a fixed axis. The spin decays as $\boldsymbol{\omega}(t)=\boldsymbol{\omega}_0\,e^{-\lambda t}$ with λ between 0 and $0.2\,\mathrm{s}^{-1}$; the spin axis keeps its direction.
Justification. The range is our assumption; even at its upper end the ball loses at most 8 % of its spin before the goal (Table 5).

A12 Knuckling as a random force. At low spin the unsteady wake exerts a random force perpendicular to the flight. Its size is calibrated to the impact scatter measured by Hong and Asai (2014), and it is switched off smoothly as the spin parameter exceeds $\mathrm{Sp}_K=0.05$ (Section 4.2).
Justification. Knuckling is usually explained by irregular vortex shedding behind a ball that does not spin; spin organises the wake. The scale Sp_K is our assumption. For the given kick the force is reduced to 0.05 ± 0.04 of its non-spinning value, so this assumption matters for low-spin kicks, not for our answer.

A13 Plausible kicks. Before the observations are used, the launch speed is taken to lie between 15 and $40\,\mathrm{m\,s}^{-1}$, a range that contains the mean penalty speeds measured in club and World Cup matches, (26 ± 3) and $(32\pm6)\,\mathrm{m\,s}^{-1}$ (Morya et al., 2005). Each component of the spin perpendicular to the flight is Gaussian with mean zero and standard deviation $\sigma_\omega=25\,\mathrm{rad\,s}^{-1}$.
Justification. Both scales are our assumption. The data disfavour $\sigma_\omega=10\,\mathrm{rad\,s}^{-1}$ against 25 by a Bayes factor of 16 and barely distinguish 25 from 50 (factor 1.8 in favour of 50). This is the assumption that matters most: Section 9 adds the evidence-weighted effect of the 25 and 50 priors to the total standard deviation of the answer.

3 Notation

Table 1 lists the symbols used in more than one section, in the order in which they appear; all others are defined where they are used. Bold symbols are vectors and the same symbol in italics denotes the magnitude ($v=|\mathbf{v}|$). A hat on a vector denotes a unit vector; on a density or covariance it denotes an estimate from samples.

4 Physical Model

We write the full flight model, M4, force by force, place it in a hierarchy of simpler models, read what the two observations say before any model is fitted, and rank the effects.

4.1 Equation of Motion

After launch (Assumptions A2 and A3) the ball feels gravity, a steady air force split into drag $\mathbf{F}_D$ along the relative velocity and Magnus force $\mathbf{F}_M$ perpendicular to it, and a fluctuating knuckling force $\mathbf{F}_K$:

$$\frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t}=\mathbf{v},\qquad m\,\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t}=-mg\,\hat{\mathbf{z}}+\mathbf{F}_D+\mathbf{F}_M+\mathbf{F}_K,\qquad \mathbf{r}(\tau)=\mathbf{r}_0, \tag{1}$$

where τ is the effective launch instant (Assumption A2) and the aerodynamic forces depend on the velocity relative to the air, $\mathbf{v}_{\mathrm{rel}}=\mathbf{v}-\mathbf{v}_{\mathrm{w}}$, with $\mathbf{v}_{\mathrm{w}}=0$ unless stated (Assumption A7). Both

Table 1: Notation.

Symbol	Meaning	Unit
x, y, z	coordinates: towards the goal, to the goalkeeper's right, up	m
L, W, H	distance to the goal plane, goal width, goal height	m
t	time since the kick	s
$\mathbf{r}(t), \mathbf{v}(t)$	position and velocity of the ball's centre	m, $\mathrm{m\,s^{-1}}$
$\mathbf{r}_0$	position at $t = 0$	m
$\mathbf{r}_1, \mathbf{r}_2$	observed positions at t_1 and t_2	m
$t_1, t_2, \Delta t$	observation times and their spacing $\Delta t = t_2 - t_1$	s
$(y^*, z^*), t^*$	entry point (crossing of $x = L$) and crossing time	m, s
r_b, A, m	ball radius, cross-section πr_b^2, mass	m, $\mathrm{m^2}$, kg
τ, σ_τ	offset of the labelled kick instant; its prior s.d. (0 here)	s
σ	standard deviation of the observation error per coordinate	m
g	gravitational acceleration	$\mathrm{m\,s^{-2}}$
ρ, ν	air density, kinematic viscosity of air	$\mathrm{kg\,m^{-3}}$, $\mathrm{m^2\,s^{-1}}$
$\mathbf{v}_\mathrm{w}$	wind velocity	$\mathrm{m\,s^{-1}}$
$\mathbf{v}_\mathrm{rel}$	velocity relative to the air, $\mathbf{v} - \mathbf{v}_\mathrm{w}$	$\mathrm{m\,s^{-1}}$
κ	aerodynamic constant $\rho A/(2m)$	$\mathrm{m^{-1}}$
$\boldsymbol{\omega}, \omega_\perp$	spin vector; its component perpendicular to $\mathbf{v}_\mathrm{rel}$	$\mathrm{rad\,s^{-1}}$
λ	spin decay rate	$\mathrm{s^{-1}}$
Re	Reynolds number $2r_b v_\mathrm{rel}/\nu$	–
Sp	spin parameter $r_b\omega_\perp/v_\mathrm{rel}$	–
C_D, C_L	drag and lift (Magnus) coefficients	–
$C_{D,\mathrm{sub}}, C_{D,\mathrm{sup}}$	drag coefficient below and above the drag crisis, without spin	–
$\mathrm{Re}_c, \Delta\mathrm{Re}$	centre and width of the drag crisis	–
$\Delta C_{D,\omega}, \mathrm{Sp}_D$	increase of C_D with spin and its saturation scale	–
$C_{L,\max}, \mathrm{Sp}_L$	plateau of C_L and the spin parameter over which it is reached	–
$\hat{\mathbf{n}}$	direction of the Magnus force	–
a_{30}	knuckling acceleration of a non-spinning ball at $30\,\mathrm{m\,s^{-1}}$	$\mathrm{m\,s^{-2}}$
$\boldsymbol{\xi}(t), \tau_c$	unit random process of the knuckling force; its correlation time	–, s
Sp_K	spin parameter above which knuckling is suppressed	–
σ_ω	prior standard deviation of each perpendicular spin component	$\mathrm{rad\,s^{-1}}$
M0–M4, CA	flight models of Table 3	–
M4-B	Bayesian predictor built on M4 (Section 5)	–
$\mathbf{d}$, $\mathbf{d}(\mathbf{u}, \boldsymbol{\phi})$	data $(\mathbf{r}_1, \mathbf{r}_2)$; the same six coordinates predicted by M4	m
$\mathbf{u}$	fitted vector $(\mathbf{v}_0, \boldsymbol{\ell}, \tau)$	mixed
$\boldsymbol{\ell}$, Sp_0	lift vector $C_L(\mathrm{Sp}_0)\,\hat{\mathbf{n}}_0$ at launch; spin parameter at launch	–
$\hat{\mathbf{e}}$	initial flight direction estimated from the data	–
$\boldsymbol{\phi}$, $\omega_\parallel$	nuisance vector (Section 5.2); spin component along $\hat{\mathbf{e}}$	mixed, $\mathrm{rad\,s^{-1}}$
$\boldsymbol{\Sigma}_d$, $\boldsymbol{\Sigma}_K$	covariance of the observation error; its knuckling part	$\mathrm{m^2}$
$p(\cdot)$, $q(\cdot)$, q_2	prior, likelihood or posterior; proposal; stage-2 proposal	–
N, N_1, N_2, N_eff	number of samples; in stage 1; in stage 2; effective sample size	–
$\mathbf{x}^*$, $\hat{p}$	crossing point (y^*, z^*); its kernel density estimate	m, $\mathrm{m^{-2}}$
R_α	highest-density region of probability α	–
$\mathbf{x}_\mathrm{true}$, η	true entry point of a test kick; its credibility level	m, –
f	calibration stretch factor, Eq. (19)	–

steady forces scale as $\frac{1}{2}\rho v_{\mathrm{rel}}^2 A$; divided by m, they carry the same constant $\kappa = \rho A/2m$, which for nominal air and ball ($\rho = 1.2\,\mathrm{kg\,m^{-3}}$, $m = 0.43\,\mathrm{kg}$) is $\kappa = 0.0530\,\mathrm{m^{-1}}$. Two dimensionless numbers control the coefficients:

$$\mathrm{Re} = \frac{2 r_b v_{\mathrm{rel}}}{\nu}, \qquad \mathrm{Sp} = \frac{r_b\,\omega_\perp}{v_{\mathrm{rel}}}, \qquad \omega_\perp = |\boldsymbol{\omega} \times \hat{\mathbf{v}}_{\mathrm{rel}}|. \tag{2}$$

Only the spin component perpendicular to the flight enters Sp: spin about the direction of flight produces no Magnus force.

4.2 Aerodynamic Forces

Drag and the drag crisis. Drag opposes the motion through the air,

$$\mathbf{F}_D = -\tfrac{1}{2}\rho A\, C_D\, v_{\mathrm{rel}}\, \mathbf{v}_{\mathrm{rel}} \quad \Longrightarrow \quad \mathbf{a}_D = -\kappa\, C_D\, v_{\mathrm{rel}}\, \mathbf{v}_{\mathrm{rel}}. \tag{3}$$

Below a critical Reynolds number Re_c the boundary layer is laminar and separates early, leaving a wide wake; above it the layer is turbulent, stays attached longer and the wake narrows, so C_D drops sharply: the drag crisis. For footballs Hong and Asai (2014) measured a fall from about 0.5 to about 0.2 or less, at an Re_c that depends on the panel shape and orientation, and trajectory analyses give $C_D \approx 0.17$ for a non-spinning ball above the crisis (Goff and Carré, 2012). We write the crisis as a smooth step and let spin raise the drag by up to $\Delta C_{D,\omega}$, an increase whose size is our assumption:

$$C_D(\mathrm{Re}, \mathrm{Sp}) = C_{D,\mathrm{sub}} - \frac{C_{D,\mathrm{sub}} - C_{D,\mathrm{sup}}}{1 + \exp[-(\mathrm{Re} - \mathrm{Re}_c)/\Delta\mathrm{Re}]} + \Delta C_{D,\omega}\left[1 - e^{-\mathrm{Sp}/\mathrm{Sp}_D}\right]. \tag{4}$$

The general method needs the crisis because it covers launch speeds of $15\,\mathrm{m\,s^{-1}}$ to $40\,\mathrm{m\,s^{-1}}$, and the slower of these kicks fly through the crisis range of Re_c. The given kick does not: its Reynolds number falls from 4.5×10^5 at t_1 to 3.6×10^5 at the goal plane (exact M3 fit, the reading of the data with the most drag), at or above the upper edge of the crisis, so its drag stays near the supercritical value; only if Re_c lies at the top of its prior range does the last part of the flight enter the crisis (Fig. 2a).

Magnus force. Spin delays separation on the side of the ball moving with the flow and advances it on the other, so the wake is deflected and the air pushes the ball sideways:

$$\mathbf{F}_M = \tfrac{1}{2}\rho A\, C_L\, v_{\mathrm{rel}}^2\, \hat{\mathbf{n}}, \qquad \hat{\mathbf{n}} = \frac{\boldsymbol{\omega} \times \mathbf{v}_{\mathrm{rel}}}{|\boldsymbol{\omega} \times \mathbf{v}_{\mathrm{rel}}|}. \tag{5}$$

Trajectory analyses measure lift coefficients of footballs at spin parameters beyond the reach of wind tunnels (Goff and Carré, 2010); C_L rises with Sp to about 0.35 at $\mathrm{Sp} \approx 0.5$ and levels off (Goff and Carré, 2012). We use the saturating form

$$C_L(\mathrm{Sp}) = C_{L,\max}\left[1 - e^{-\mathrm{Sp}/\mathrm{Sp}_L}\right]. \tag{6}$$

Equation (5) has a removable singularity: $\hat{\mathbf{n}}$ is undefined when the ball does not spin. Using $|\boldsymbol{\omega} \times \mathbf{v}_{\mathrm{rel}}| = \omega_\perp v_{\mathrm{rel}} = \mathrm{Sp}\, v_{\mathrm{rel}}^2/r_b$ from Eq. (2),

$$\mathbf{a}_M = \frac{\mathbf{F}_M}{m} = \kappa\, r_b\, \frac{C_L(\mathrm{Sp})}{\mathrm{Sp}}\, \boldsymbol{\omega} \times \mathbf{v}_{\mathrm{rel}}. \tag{7}$$

The ratio C_L/Sp tends to $C_{L,\max}/\mathrm{Sp}_L$ as $\mathrm{Sp} \to 0$, so $\mathbf{a}_M$ is a smooth function of $\boldsymbol{\omega}$ that vanishes linearly with the spin; smoothness matters because the inference of Section 5 fits the spin by gradient steps. The two limits are physical checks: for small spin, the classical linear Magnus law $\mathbf{a}_M \propto \boldsymbol{\omega} \times \mathbf{v}_{\mathrm{rel}}$; for large spin, $|\mathbf{a}_M| \to \kappa C_{L,\max} v_{\mathrm{rel}}^2$, independent of the spin rate. Figure 2 shows both laws over the prior ranges of Table 2, with the given kick placed on them.

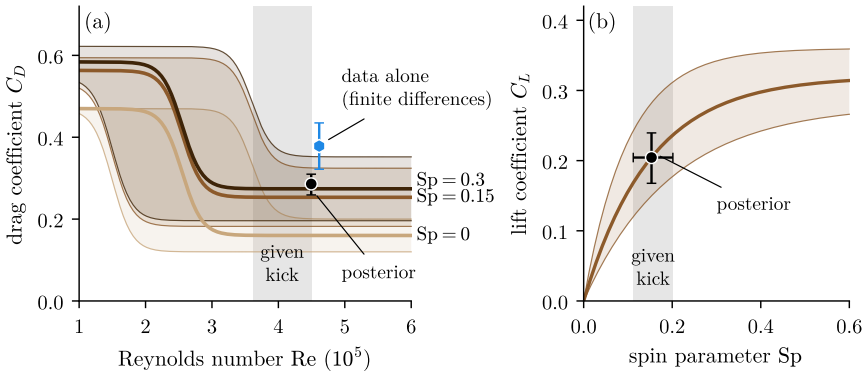


Figure 2: Drag and lift laws over the prior ranges (shaded). The given kick (grey band: its Reynolds numbers along the flight) flies above the drag crisis or at its upper edge; its posterior drag coefficient (black) lies inside the prior band, and the data-alone value from finite differences (blue, reproduced by the exact M3 fit) lies above it, but by less than one standard error: the mild tension of Section 4.4. Its spin parameter sits on the rising part of the lift curve.

Spin decay. Air friction slowly removes spin; we take $\boldsymbol{\omega}(t) = \boldsymbol{\omega}_0\, e^{-\lambda t}$ about a fixed axis with $0 \le \lambda \le 0.2\,\mathrm{s}^{-1}$ (Assumption A11), so the ball loses at most 8 % of its spin before the goal.

Knuckling. A ball with little spin is pushed sideways by fluctuating forces: in wind-tunnel tests Hong and Asai (2014) measured side and lift forces with standard deviations of 0.6 N to 3.1 N, and non-spinning kicks from their robot scattered by 0.13 m to 0.68 m at 25 m. We model this as a random acceleration perpendicular to the flight,

$$\mathbf{a}_K(t) = a_{30} \left(\frac{v_{\mathrm{rel}}}{30\,\mathrm{m\,s^{-1}}} \right)^2 e^{-\mathrm{Sp}/\mathrm{Sp}_K} \left[\boldsymbol{\xi}(t) - \left(\boldsymbol{\xi}(t) \cdot \hat{\mathbf{v}}_{\mathrm{rel}} \right) \hat{\mathbf{v}}_{\mathrm{rel}} \right], \tag{8}$$

where each component of $\boldsymbol{\xi}$ is an independent Ornstein–Uhlenbeck process (Uhlenbeck and Ornstein, 1930) with unit variance and correlation time τ_c, and $e^{-\mathrm{Sp}/\mathrm{Sp}_K}$ switches the force off for a spinning ball (Assumption A12). We fix a_{30} by simulating non-spinning kicks at $30\,\mathrm{m\,s^{-1}}$ and scaling it until their scatter at 25 m is 0.30 m, the middle of the measured range (Appendix A.2). For a non-spinning penalty, knuckling then scatters the entry point by 0.087 m, comparable to the uncertainty of our prediction, so the general method keeps it; for the given kick the spin reduces the force to 0.05 ± 0.04 of that, which is negligible.

Parameters. Table 2 lists every parameter of M4 with its value or range and its source. The ranges express Assumption A10: they span the balls and conditions the method should cover, and Section 5 turns them into priors. None is tuned to the given kick.

4.3 A Hierarchy of Models

A good model is the simplest one the data cannot reject, so we compare M4 with the simpler models of Table 3, obtained by switching effects off, and test all of them in Section 7. All are integrated with the classical fourth-order Runge–Kutta method (Press et al., 2007) with a step of 2 ms (2.5 ms up to t_2); a step eight times smaller moves the entry point by less than 10^{-12} m (Appendix A.1). M3 already fits the six data exactly; we go further because M3 holds C_D and C_L fixed while the ball slows, and an exact fit cannot say how wrong it might be. M4 has more unknowns than data; the priors of Table 2 supply the rest, and their spread is part of the uncertainty.

Table 2: Parameters of the full model M4; ranges are uniform priors, s.d. are Gaussian. "Assumption" marks a value we chose; it is varied in Section 9 where it matters.

Quantity	Symbol	Value or range	Source
Ball radius	r_b	0.11 m	problem; IFAB, 2026
Ball mass	m	0.41–0.45 kg	IFAB, 2026
Air density	ρ	1.05–1.25 $\mathrm{kg\,m^{-3}}$	assumption (A8)
Kinematic viscosity	ν	$1.54 \times 10^{-5}\ \mathrm{m^2\,s^{-1}}$	Goff and Carré (2012)
Drag below the crisis	$C_{D,\mathrm{sub}}$	0.47	assumption, from Hong and Asai (2014)
Drag above the crisis	$C_{D,\mathrm{sup}}$	0.12–0.20	Hong and Asai, 2014; Goff and Carré, 2012
Centre of the crisis	Re_c	$1.5\text{–}3.6 \times 10^5$	Hong and Asai (2014), extended down (A10)
Width of the crisis	$\Delta\mathrm{Re}$	1.5×10^4	assumption
Drag increase with spin	$\Delta C_{D,\omega}$	0.08–0.16	assumption
Saturation of the increase	Sp_D	0.10	assumption
Lift plateau	$C_{L,\max}$	0.28–0.36	assumption, around Goff and Carré (2012)
Rise of lift	Sp_L	0.10–0.20	assumption
Spin decay rate	λ	0–0.2 $\mathrm{s^{-1}}$	assumption (A11)
Knuckling scatter at 25 m	–	0.30 m	Hong and Asai (2014), mid-range
Knuckling correlation time	τ_c	0.06 s	assumption (Appendix A.2)
Knuckling suppression	Sp_K	0.05	assumption
Launch speed	v_0	15–40 $\mathrm{m\,s^{-1}}$	assumption; contains Morya et al. (2005)
Perpendicular spin, s.d.	σ_ω	25 $\mathrm{rad\,s^{-1}}$	assumption (A13)
Launch offset, s.d.	σ_τ	0; 2 ms for tracking data	problem; assumption (A2)
Observation error	σ	2.9 mm (rounding)	problem (A4)

Table 3: Flight models. "Fitted" counts the unknowns determined from the six observed coordinates.

Model	Forces	Fitted	Use in this paper
M0	none: straight-line extrapolation of the launch velocity	3 ($\mathbf{v}_0$)	baseline
M1	gravity: parabola	3 ($\mathbf{v}_0$)	baseline
CA	constant acceleration (no physics)	6 ($\mathbf{v}_0, \mathbf{a}$)	baseline
M2	gravity and drag with constant C_D	4 ($\mathbf{v}_0, C_D$)	baseline
M3	M2 and Magnus force, constant C_L and direction	6 ($\mathbf{v}_0, C_D$, 2 Magnus comp.)	exact fit; delta method (Casella and Berger, 2002)
M4	gravity, $C_D(\mathrm{Re}, \mathrm{Sp})$, $C_L(\mathrm{Sp})$, spin decay, knuckling; optional wind (zero in the headline, A7)	$\mathbf{v}_0$, 2 spin comp., τ (rest: priors)	Bayesian (Sec. 5)

4.4 What the Two Observations Already Say

Before fitting any model, three positions equally spaced in time give the velocity and acceleration at t_1 by central differences,

$$\mathbf{v}(t_1) \approx \frac{\mathbf{r}_2 - \mathbf{r}_0}{2\Delta t}, \qquad \mathbf{a}(t_1) \approx \frac{\mathbf{r}_2 - 2\mathbf{r}_1 + \mathbf{r}_0}{\Delta t^2}, \tag{9}$$

which are exact for constant acceleration and a launch from $\mathbf{r}_0$ at exactly $t = 0$. Since $\mathbf{r}_0$ is exact and $\mathbf{r}_1$, $\mathbf{r}_2$ carry independent errors σ, each component of $\mathbf{a}$ has standard deviation $\sqrt{5}\,\sigma/\Delta t^2 = 2.6\,\mathrm{m\,s^{-2}}$. For the given kick $v(t_1) = (31.461 \pm 0.029)\,\mathrm{m\,s^{-1}}$ and $\mathbf{a}(t_1) = (-16.0,\ -16.0,\ -12.0)\,\mathrm{m\,s^{-2}}$. The launch speed is $32.6\,\mathrm{m\,s^{-1}}$, within the $(32 \pm 6)\,\mathrm{m\,s^{-1}}$ measured for World Cup penalties (Morya et al., 2005), at $12\,^\circ$ above the horizontal and $17\,^\circ$ towards the goalkeeper's right. Splitting the aerodynamic acceleration $\mathbf{a} + g\hat{\mathbf{z}}$ along and perpendicular to $\hat{\mathbf{v}}(t_1)$ (uncertainties combine rounding with the ranges of ρ and m):

1. **Drag.** The ball decelerates along its path at $(19.9 \pm 2.6)\,\mathrm{m\,s^{-2}}$, an effective $C_D = |a_\parallel|/(\kappa v^2) = 0.38 \pm 0.06$, well above the 0.12–0.20 of a non-spinning ball beyond the crisis; the paragraph below shows how this reading depends on the launch instant.
2. **Sideways force.** The perpendicular acceleration is $(11.1 \pm 2.5)\,\mathrm{m\,s^{-2}}$, more than four standard errors. With gravity removed and drag along the path, it needs a Magnus force or a knuckling force of 4.7 times its calibrated standard deviation (Eq. (8) with $\mathrm{Sp} = 0$): spin is the explanation by a wide margin, with $C_L = |a_\perp|/(\kappa v^2) = 0.21 \pm 0.05$.
3. **Spin.** Inverting Eq. (6) with mid-range lift parameters gives $\mathrm{Sp} \approx 0.16$ and $\omega_\perp \approx 46\,\mathrm{rad\,s^{-1}}$ (68 % interval over the lift ranges: 30–86 $\mathrm{rad\,s^{-1}}$).
4. **Spin axis.** The Magnus force points mainly towards $-y$, back towards the middle of the goal, so the spin axis $\hat{\mathbf{v}} \times \hat{\mathbf{n}}$ lies within $(15 \pm 7)\,^\circ$ of pointing straight down: mostly sidespin with a little backspin.

The launch instant and the drag. More drag and a kick a fraction of a millisecond earlier are indistinguishable in two observations; the entry point is the same. The problem states the kick instant exactly, and the headline takes it so ($\tau = 0$, Assumption A2). For tracking data whose kick is labelled only to a camera frame, the method also accepts an offset τ of either sign. If free flight starts from $\mathbf{r}_0$ at $t = \tau$ with velocity $\mathbf{v}$ and nearly constant acceleration $\mathbf{a}$, then

$$\mathbf{r}_2 - 2\mathbf{r}_1 + \mathbf{r}_0 = \mathbf{a}\,\Delta t^2 + \mathbf{v}\,\tau - \tfrac{1}{2}\mathbf{a}\,\tau^2 \approx \mathbf{a}\,\Delta t^2 + \mathbf{v}\,\tau. \tag{10}$$

An offset adds $\mathbf{v}\tau/\Delta t^2$ to the measured acceleration along the flight: it mimics a change of drag but not the perpendicular acceleration, so the evidence for spin stands. Read with $\tau = 0$, the data prefer the drag of the first rows of Table 4, above the cap of our drag prior, $C_{D,\mathrm{sup}} + \Delta C_{D,\omega}(1 - e^{-\mathrm{Sp}/\mathrm{Sp}_D})$ with both at their upper limits, by less than one standard error. With the instant exact (headline), the data pull the drag into the upper part of its prior. With the instant free, a launch $(-0.59 \pm 0.25)\,\mathrm{ms}$ early supplies $7.5\,\mathrm{m\,s^{-2}}$ of apparent deceleration, about the gap $6.9\,\mathrm{m\,s^{-2}}$ between the model-free and the posterior drag, and C_D returns to the middle of its prior. A physical contact delay, $\tau \geq 0$, would mimic *less* deceleration, the opposite of what the data ask for, so it would pile up at $\tau = 0$ and coincide with the headline. The data constrain launch instant and drag together, and the drag prior selects the offset; the entry points of the two readings differ by less than twice the Monte Carlo noise (Section 9); the crossing time changes little and the perpendicular spin, $(45 \pm 13)\,\mathrm{rad\,s^{-1}}$ against $(38 \pm 13)\,\mathrm{rad\,s^{-1}}$, stays within its spread.

Table 4: The launch instant and the drag: the readings differ in drag, not in the entry point. C_D is the effective value at t_1 (posterior median ± half the 68 % interval); entry points are mode ± s.d. The first two rows read the data alone, with the instant exact; the last two are sensitivity readings.

Reading	τ	C_D	y^*	z^*	t^*
Finite differences	0	0.38 ± 0.06	–	–	–
Exact M3 fit	0	0.378	–	–	–
Instant exact (headline)	0	0.286 ± 0.025	(2.67 ± 0.10) m	(1.97 ± 0.13) m	(0.392 ± 0.003) s
Instant free	(-0.59 ± 0.25) ms	0.25 ± 0.03	(2.70 ± 0.11) m	(1.96 ± 0.12) m	(0.390 ± 0.003) s
Contact delay, $\tau \geq 0$	$\to 0$	as the headline			

4.5 Which Effects Matter

Table 5 ranks the effects for the given kick: hand estimates of the acceleration and of the displacement $\frac{1}{2} a\, t_f^2$ over the constant-acceleration flight time $t_f \approx 0.41$ s, and simulated shifts with the effect switched off or on around the exact M3 fit (spin decay and wind: around the M4 posterior mean). Gravity and the Magnus force each move the entry point by most of a metre; drag and a change of the air density by about one standard deviation of the prediction; every other effect by less. Gravity, the Magnus force and drag act far above the precision of the measured acceleration, and models without all three (M0–M2) fail (Section 7). The air and the ball enter as prior ranges, wind as a mismatch test, and the rest is negligible for this kick.

Table 5: Effects on the given kick; a shift is the distance between entry points with and without the effect, with launch velocity and coefficients held fixed; when the kick is re-inferred from the observations, most of the air-density and wind effect is absorbed (Section 9: 0.017 m and 0.036 m). † Shift (simulated where available) smaller than the smaller standard deviation of the prediction, 0.10 m in y^*.

Effect	Acceleration	Shift, estimate	Shift, simulated	Treatment
Gravity	$9.81\,\mathrm{m\,s^{-2}}$	0.80 m	0.73 m	modelled
Magnus force	$(11.1 \pm 2.5)\,\mathrm{m\,s^{-2}}$	0.91 m	0.77 m	modelled, spin inferred
Drag	$(19.9 \pm 2.6)\,\mathrm{m\,s^{-2}}$	–	0.11 m	modelled, $C_D(\mathrm{Re}, \mathrm{Sp})$
Air density −16 %	–	–	0.12 m	prior range
Ball mass +5 %†	–	–	0.035 m	prior range
Spin decay, $\lambda = 0.2\,\mathrm{s^{-1}}$†	–	–	0.010 m	prior range
Crosswind $2\,\mathrm{m\,s^{-1}}$†	$1.3\,\mathrm{m\,s^{-2}}$	0.10 m	0.081 m	mismatch test; Sec. 9
Buoyancy†	$0.15\,\mathrm{m\,s^{-2}}$	13 mm	0.011 m	neglected (A6)
Knuckling (this kick)†	$0.09\,\mathrm{m\,s^{-2}}$	$\lesssim 8$ mm	–	random force
Coriolis†	$0.005\,\mathrm{m\,s^{-2}}$	0.4 mm	–	neglected (A5)

5 Prediction Method

5.1 The Idea

Imagine many kicks, each with its own launch velocity, spin, ball and air, drawn from the priors. Weighting each by how closely it passes $\mathbf{r}_1$ and $\mathbf{r}_2$ and flying it on to $x = L$ gives a weighted cloud of crossing points that, by Bayes' theorem, samples the predictive density $p(y^*, z^* \mid \mathbf{r}_1, \mathbf{r}_2)$: its densest point is the prediction, its spread the uncertainty (Figure 3). As the data fix the speed at t_1 to $\pm 0.029\,\mathrm{m\,s^{-1}}$, blind draws almost never pass close to both observations, so we fit the few unknowns the data measure well, draw the rest from their priors, and correct with importance weights.

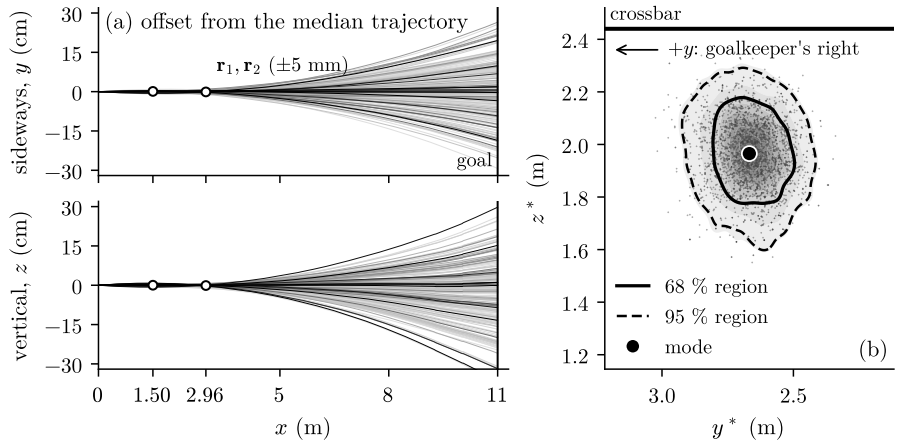


Figure 3: The method for the given kick. (a) Candidate trajectories fitted through the two observations (circles, ±5 mm rounding), drawn as their sideways and vertical offset in centimetres from the weighted median trajectory; darker lines carry larger weight. (b) Their crossing points on the goal plane, seen from the penalty mark as in Figure 6 (+y, the goalkeeper's right, to the left; equal scales), with the density estimate (shading), the 68 % and 95 % regions and the mode. Trajectories that agree with the data to millimetres spread by decimetres at the goal: that spread is the uncertainty.

5.2 Unknowns and Priors

The unknowns the six data $\mathbf{d} = (\mathbf{r}_1, \mathbf{r}_2)$ determine well are fitted; everything else is averaged over its prior. The *fitted vector* is

$$\mathbf{u} = (\mathbf{v}_0,\ \boldsymbol{\ell},\ \tau), \qquad \boldsymbol{\ell} = C_L(\mathrm{Sp}_0)\,\hat{\mathbf{n}}_0, \tag{11}$$

the launch velocity, the *lift vector* at launch (Sp_0, $\hat{\mathbf{n}}_0$ as in Eq. (5)), perpendicular to the initial flight direction $\hat{\mathbf{e}}$ estimated by the three-point formula, and the offset τ of the labelled kick instant from the observation clock (Assumption A2). For tracking data whose kick instant is labelled only to a camera frame, τ may have either sign and gets the prior $\mathcal{N}(0, \sigma_\tau^2)$ with $\sigma_\tau = 2\,\mathrm{ms}$; the given kick states the instant exactly, so its prediction uses $\sigma_\tau = 0$, which fixes $\tau = 0$ and drops it from $\mathbf{u}$ (Section 4.4 shows that the two readings give the same entry point). The *nuisance vector* $\boldsymbol{\phi} = (C_{D,\mathrm{sup}}, \Delta C_{D,\omega}, \mathrm{Re}_c, C_{L,\max}, \mathrm{Sp}_L, \rho, m, \lambda, \omega_\parallel)$ holds the other parameters of M4; $\omega_\parallel$, the spin component along $\hat{\mathbf{e}}$, produces no Magnus force at launch. The first eight are uniform on the ranges of Table 2 (Assumption A10); the launch speed is flat on $15\,\mathrm{m\,s^{-1}}$ to $40\,\mathrm{m\,s^{-1}}$, and each Cartesian spin component is $\mathcal{N}(0, \sigma_\omega^2)$ (Assumption A13). We fit $\boldsymbol{\ell}$ rather than $\boldsymbol{\omega}_\perp$ because the sideways acceleration at launch, $\kappa v_0^2 \boldsymbol{\ell}$ (Eq. (7)), is linear in $\boldsymbol{\ell}$, while near the plateau of C_L the posterior of $\boldsymbol{\omega}_\perp$ is a curved ridge that defeats a Gaussian approximation; the Jacobian of this change of variables carries the spin prior over exactly (Appendix B.1).

5.3 Algorithm

M4-B is self-normalised importance sampling (Owen, 2013) of the joint posterior

$$p(\mathbf{u}, \boldsymbol{\phi} \mid \mathbf{d}) \propto p(\mathbf{d} \mid \mathbf{u}, \boldsymbol{\phi})\, p(\mathbf{u} \mid \boldsymbol{\phi})\, p(\boldsymbol{\phi}), \tag{12}$$

with the Laplace approximation (Tierney and Kadane, 1986) $q(\mathbf{u} \mid \boldsymbol{\phi}) = \mathcal{N}(\bar{\mathbf{u}}, \boldsymbol{\Gamma}^{-1})$ as the proposal for $\mathbf{u}$. Each $\mathbf{u}_i$ is *drawn* from q, not set to its best fit $\bar{\mathbf{u}}_i$: the crossing points then sample the predictive density integrated over $\mathbf{u}$ and $\boldsymbol{\phi}$, whereas flying only $\bar{\mathbf{u}}_i$ would drop the spread of the kick that the rounding allows, the largest term here (Section 6.4). In outline: (1) draw nuisance parameters from their priors; (2) for each, fit the kick to the data and draw a

kick around that fit; (3) weight each draw by how well it explains the data, divided by how likely it was to be drawn; (4) fly each kick to the goal; (5) smooth the weighted crossing points into a density and read off mode and regions. The box below gives the details. The likelihood is Gaussian,

$$p(\mathbf{d} \mid \mathbf{u}, \boldsymbol{\phi}) \propto \frac{e^{-\chi^2/2}}{\sqrt{\det \boldsymbol{\Sigma}_d}}, \quad \chi^2 = \boldsymbol{\delta}^\top \boldsymbol{\Sigma}_d^{-1} \boldsymbol{\delta}, \quad \boldsymbol{\delta} = \mathbf{d} - \mathbf{d}(\mathbf{u}, \boldsymbol{\phi}), \quad \boldsymbol{\Sigma}_d = \sigma^2 \mathbf{I}_6 + \boldsymbol{\Sigma}_K, \tag{13}$$

with $\mathbf{d}(\mathbf{u}, \boldsymbol{\phi})$ the positions at t_1, t_2 of the M4 flight without knuckling and $\boldsymbol{\Sigma}_K$ the covariance of the knuckling push in the observation window, negligible for a spinning ball (Appendix B.4). It replaces the ± 5 mm rounding box by a Gaussian of the same standard deviation (Assumption A4), which costs nothing measurable (Sections 7.3 and 7.5).

Algorithm M4-B. Input $\mathbf{d}$, σ, budget $N = N_1 + N_2$. Output weighted crossing points.

1. Draw N_1 nuisance vectors $\boldsymbol{\phi}_i$ from the prior $p(\boldsymbol{\phi})$.
2. For each $\boldsymbol{\phi}_i$, fit $\mathbf{u}$ by Gauss–Newton (Nocedal and Wright, 2006) on the nine residuals $\mathbf{R} = (\boldsymbol{\Sigma}_d^{-1/2}\boldsymbol{\delta},\ \boldsymbol{\omega}_\perp(\boldsymbol{\ell})/\sigma_\omega,\ \tau/\sigma_\tau)$: start at the three-point $\mathbf{v}_0$, $\boldsymbol{\ell} = 0$, $\tau = 0$; step $-(\mathbf{J}^\top\mathbf{J})^{-1}\mathbf{J}^\top\mathbf{R}$ ($\mathbf{J}$ by finite differences), keeping $\boldsymbol{\ell}$ below the plateau (with $\sigma_\tau = 0$: $\tau = 0$, eight residuals). Result: $\bar{\mathbf{u}}_i$, $\boldsymbol{\Gamma}_i = \mathbf{J}^\top\mathbf{J}$.
3. Draw $\mathbf{u}_i \sim \mathcal{N}(\bar{\mathbf{u}}_i, \boldsymbol{\Gamma}_i^{-1})$, compute $\mathbf{d}(\mathbf{u}_i, \boldsymbol{\phi}_i)$, and fly to $x = L$ with a fresh knuckling realisation: crossing point $\mathbf{x}_i^* = (y_i^*, z_i^*)$ and time t_i^*, or ground contact.
4. Weight by Eq. (14) with $q = p$; effective sample size $N_{\text{eff}} = (\sum_i \tilde{w}_i)^2 / \sum_i \tilde{w}_i^2$.
5. Draw N_2 more vectors $\boldsymbol{\phi}_i$: from the prior if $N_{\text{eff}} \geq N_1/2$, else from an adapted Gaussian q_2 (Appendix B.2); repeat Steps 2–3 for them.
6. Final weight of every sample of both stages (Appendix B.3):

$$\tilde{w}_i = \frac{p(\mathbf{d} \mid \mathbf{u}_i, \boldsymbol{\phi}_i)\, p(\mathbf{u}_i \mid \boldsymbol{\phi}_i)\, p(\boldsymbol{\phi}_i)}{q(\mathbf{u}_i \mid \boldsymbol{\phi}_i)\, q(\boldsymbol{\phi}_i)}, \qquad q(\boldsymbol{\phi}) = \tfrac{N_1}{N}\, p(\boldsymbol{\phi}) + \tfrac{N_2}{N}\, q_2(\boldsymbol{\phi}). \tag{14}$$

The squared length of $\mathbf{R}$ is $-2 \ln p(\mathbf{u} \mid \mathbf{d}, \boldsymbol{\phi})$ up to a constant and the log-Jacobian of the spin prior, which the weight restores through $p(\mathbf{u} \mid \boldsymbol{\phi}) = p(\boldsymbol{\ell} \mid \mathbf{v}_0, \boldsymbol{\phi})\, \mathcal{N}(\tau; 0, \sigma_\tau^2)$ (Eq. (22); the last factor is absent when $\sigma_\tau = 0$), zero for speeds outside $15\,\mathrm{m\,s^{-1}}$ to $40\,\mathrm{m\,s^{-1}}$ or $\boldsymbol{\ell}$ at the plateau; if Step 5 draws from the prior, $q(\boldsymbol{\phi}) = p(\boldsymbol{\phi})$. N_{eff} is the usual effective sample size (Kong, 1992). Normalising the weights of the kicks that reach the goal, $w_i = \tilde{w}_i / \sum_j \tilde{w}_j$, gives the kernel density estimate

$$\hat{p}(\mathbf{x}) = \sum_i w_i\, \mathcal{N}(\mathbf{x}; \mathbf{x}_i^*, \mathbf{B}), \qquad \mathbf{B} = \hat{\boldsymbol{\Sigma}}\, N_{\text{eff}}^{-1/3}, \tag{15}$$

with $\hat{\boldsymbol{\Sigma}}$ the weighted covariance of the crossing points (Scott's rule (Scott, 1992) with N_{eff} points). The *mode* is the maximum of $\hat{p}$ on a grid of ± 4 standard deviations about the mean, refined by mean shift (Comaniciu and Meer, 2002): a point moves to the kernel-weighted average of nearby samples until it stops. The region R_α of probability α is where $\hat{p}$ exceeds its $(1-\alpha)$ quantile at fresh draws from $\hat{p}$ (the samples themselves would count their own kernels and shrink the regions).

Settings (inputs, the same for every kick): $\sigma_\tau = 0$ for the given kick and its matched tests, $\sigma_\tau = 2$ ms otherwise; $N = 20000$ for the given kick, 8000 per sensitivity run, 4000 per variance-decomposition run and 1200 per test kick; $N_1 = \max(0.3N, 300)$; at most six Gauss–Newton iterations (four in tests) with $|\boldsymbol{\ell}| \leq 0.98\, C_{L,\max}$; widening 1.5 for q_2; density grid 160^2 (100^2 in tests, 120^2 in the decomposition); 6000 draws for R_α. They affect only the Monte Carlo noise: over four seeds the mode of the given kick scatters by 0.021 m at $N = 20000$ and 0.028 m at $N = 8000$, and $N_{\text{eff}} = 4100$. The code uses NumPy (Harris et al., 2020), SciPy (Virtanen et al., 2020) for the chi-squared quantiles of the Gaussian ellipses, and matplotlib (Hunter, 2007) for the figures.

5.4 Alternative Predictors

Ten alternatives (Table 6) test whether each ingredient of M4-B is worth its cost. The *physical predictors* of Table 3 are fitted to the data and flown to the goal in milliseconds or less; M3, solved exactly by Newton's method, is the most physics the data determine on their own, and the delta method (DM) adds its rounding uncertainty at the cost of seven M3 fits (Section 6.3).

Table 6: Predictors compared in Section 7, with the time per kick: mean per prediction on one CPU core (M0–M3 in batches of 120 kicks, learned methods after training); for M4-B the median at the test budget $N = 1200$, with two test runs sharing two cores.

Predictor	Physics used	Fitted or learned	Uncertainty	Time
M0 straight line	none	$\mathbf{v}_0$ (three-point)	none	3.0×10^{-4} ms
M1 gravity only	gravity	$\mathbf{v}_0$ (least squares)	none	5.9×10^{-4} ms
CA constant accel.	none	$\mathbf{v}_0$, $\mathbf{a}$ (exact)	none	3.2×10^{-4} ms
M2 drag only	gravity, constant C_D	$\mathbf{v}_0$, C_D (least squares)	none	1.3 ms
M3 exact spin fit	M2, Magnus, constant C_L	$\mathbf{v}_0$, C_D, Magnus (exact)	none	5.1 ms
DM delta method	as M3	as M3	Gaussian (rounding)	7× M3
M4-B Bayesian	full M4	$\mathbf{u}$ fitted, $\boldsymbol{\phi}$ sampled	full density	1700 ms
GRID3, GRID5	M4 simulations	table of entry ninths	ninth probabilities	0.0021 ms
COND conditioning	M4 simulations	kernel width	kernel density	320 ms
CGMM mixture	M4 simulations	80-component mixture	Gaussian mixture	90 ms

Learned predictors approximate the same posterior predictive density without an equation of motion. Trained on 200 000 kicks from the generator of Section 7.1 (M4, our priors, separate seed), they know no more than M4-B and share its blind spots; their value is speed, which makes them the deployable version of the method: the mixture needs 90 ms per prediction and M4-B 1700 ms at the test budget $N = 1200$ and 17 s at $N = 20000$, while a goalkeeper has 0.29 s after the second observation. The two M4-B times are not in proportion to N: they were measured under different load (two test runs shared two cores; the headline prediction ran alone), and all samples are fitted and flown together as arrays, so the fixed cost of each step is shared by more samples at larger N. *Grid lookup (GRID3)*, the model-free baseline, extends each ninth of the goal back to the mark as a pyramid, counts on the training kicks how often each pair of pyramids (or "outside") holding $\mathbf{r}_1$, $\mathbf{r}_2$ leads to each entry ninth, plus one, and predicts the centre of the most probable ninth (GRID5: 5×5 pyramids). It sees the direction of the kick, not its bend: the given kick climbs above the line from mark to crossbar, so both observations lie outside every pyramid, and the grid picks a ninth on the wrong side of the goal ($y^* = (-2.4 \pm 2.2)$ m). *Dataset conditioning (COND)* weights each training kick j by $\exp(-|\mathbf{d}_j - \mathbf{d}|^2/2h^2)$ and forms Eq. (15) from the weighted crossing points, with $h = 3$ cm chosen among 1, 2, 3 and 5 cm by held-out log score; in six dimensions few stored kicks lie that close (median effective sample size 39). *The conditional Gaussian mixture (CGMM)* fits 20, 40 or 80 full-covariance Gaussians (chosen by held-out likelihood) to $(\mathbf{r}_1, \mathbf{r}_2, y^*, z^*)$ of the training kicks (Pedregosa et al., 2011), conditions each on $\mathbf{d}$ in closed form, reweights it by its marginal density at $\mathbf{d}$, and takes the mode of this mixture on a grid.

5.5 Inputs, Outputs and Domain of Validity

Inputs: $\mathbf{r}_1$, $\mathbf{r}_2$, σ. *Outputs:* mode, 68 % and 95 % regions, mean, standard deviations, probabilities of reaching the goal plane, entering cleanly and striking the frame, crossing time, and the posterior of the kick. *Domain:* kicks from the mark at 15 m s^{-1} to 40 m s^{-1}, balls and air inside Table 2, spins up to a few σ_ω, still air, no contact before the observations; nothing is tuned to the given kick.

6 Uncertainty Estimation

The uncertainty is the spread of the predictive density, which carries every modelled source at once; a linearised calculation checks its size and Section 7 its honesty. For the given kick the standard deviations (s.d.) are 0.10 m in y^* and 0.13 m in z^*. The rounding of the two observations sets their scale; the unknown ball and air add 28 % of the variance in y^* and nothing measurable in z^* (Section 6.4).

6.1 Sources of Uncertainty

Every source we know of is fitted, sampled, or tested as a deliberate mismatch (Table 7); wind and errors in the form of the model cannot be inferred from two observations, so the mismatch test measures their effect (Section 7.4).

Table 7: Sources of uncertainty and how each enters: "fitted" to the data, or "sampled" from the prior and weighted.

Source	Parameters	How it enters
Rounding of the data	$\sigma = 2.9\,\mathrm{mm}$	Gaussian likelihood, Eq. (13) (A4)
Launch velocity, timing	$\mathbf{v}_0$, τ	fitted in $\mathbf{u}$; flat speed prior; $\tau = 0$ for the given kick
Spin perpendicular to flight	$\boldsymbol{\ell}$, i.e. $\boldsymbol{\omega}_\perp$	fitted in $\mathbf{u}$; Gaussian spin prior via Eq. (22)
Spin along the flight	$\omega_\parallel$	sampled in $\boldsymbol{\phi}$, Gaussian prior
Drag and lift curves	$C_{D,\mathrm{sup}}$, $\Delta C_{D,\omega}$, Re_c, $C_{L,\max}$, Sp_L	sampled in $\boldsymbol{\phi}$, uniform on the ranges of Table 2
Air, mass, spin decay	ρ, m, λ	sampled in $\boldsymbol{\phi}$, uniform
Knuckling	$\boldsymbol{\xi}(t)$	$\boldsymbol{\Sigma}_K$ in the likelihood; one realisation per kick to the goal
Numerical integration	step size	convergence check: $4 \times 10^{-13}\,\mathrm{m}$ (Section A.1)
Wind	$\mathbf{v}_\mathrm{w}$	not modelled (A7); mismatch test only
Form of the model	coefficient curves	not modelled; mismatch test only

6.2 From Samples to Regions

The predictive density integrates the entry point over the joint posterior of Eq. (12) and over the knuckling force,

$$p(\mathbf{x} \mid \mathbf{d}) = \int \delta(\mathbf{x} - \mathbf{x}^*(\mathbf{u}, \boldsymbol{\phi}, \boldsymbol{\xi}))\, p(\boldsymbol{\xi} \mid \mathbf{u}, \boldsymbol{\phi})\, p(\mathbf{u}, \boldsymbol{\phi} \mid \mathbf{d})\, \mathrm{d}\boldsymbol{\xi}\, \mathrm{d}\mathbf{u}\, \mathrm{d}\boldsymbol{\phi}, \tag{16}$$

where $\mathbf{x}^*(\cdot)$ is the crossing point of that flight. The weighted crossing points of Section 5.3 sample this density, so every source contributes in proportion to its real effect, with correlations and the non-linearity of the flight; nothing is added in quadrature. We summarise it by highest-density regions R_α, the smallest sets holding probability α; unlike a Gaussian ellipse, they bend with a curved ridge and are cut off where the ball would hit the ground. For the given kick the half-widths of $R_{0.68}$ are 0.15 m in y^* and 0.20 m in z^*, those of $R_{0.95}$ 0.25 m and 0.35 m. The problem asks for the mode, which differs from the mean when the density is skewed; here the mode $(2.67, 1.97)\,\mathrm{m}$ and the mean $(2.66, 1.97)\,\mathrm{m}$ lie well inside each other's 68 % region.

6.3 Independent Check: the Delta Method

The delta method (Casella and Berger, 2002) propagates the rounding error through the exact fit of M3, $\mathbf{d} \mapsto \mathbf{x}^*_{\mathrm{M3}}(\mathbf{d})$, by linearisation, with no priors and no sampling. For data errors with

covariance $\sigma^2\mathbf{I}_6$ the entry point is Gaussian with covariance

$$\mathbf{\Sigma}_{\mathrm{DM}} = \sigma^2\,\mathbf{G}\,\mathbf{G}^{\top}, \qquad G_{jk} = \frac{\partial x^*_{\mathrm{M3},j}}{\partial d_k}, \tag{17}$$

where each column of the 2×6 Jacobian $\mathbf{G}$ comes from repeating the exact fit with one coordinate of $\mathbf{d}$ moved by 0.1 mm. It contains only the rounding, amplified over the rest of the flight, and leaves out everything M3 holds fixed: the nuisance parameters and the difference between M3's constant coefficients and the true flight. For the given kick it centres on (2.64 m, 1.96 m) with s.d. 0.13 m and 0.13 m, comparable with 0.0866 m and 0.126 m for M4-B with every source but the rounding removed (Table 8): equal in z^*, wider in y^*, where the spin prior and coefficient curves of M4-B change how strongly a millimetre of rounding is amplified. Holding everything else fixed, it is over-confident on the test kicks (68 % and 95 % ellipses cover 59 % and 88 %, Section 7.3): it checks the size of the dominant term, not the whole uncertainty.

6.4 Which Uncertainty Dominates

The rounding alone, with every other input fixed at mid-range and no knuckling, gives s.d. 0.0866 m in y^* and 0.126 m in z^*; adding every prior gives 0.102 m and 0.125 m (Table 8, $N = 4000$ per run, one seed for all runs). To tell effects from sampling noise we repeated the full run with three more seeds. The Monte Carlo resolution of a share is twice the standard deviation over the four seeds of $1-(\mathrm{s.d.}_{\mathrm{seed}}/\overline{\mathrm{s.d.}})^2$, the share by which the full method's own variance scatters; it is 15 % (larger axis), and the table shows only shares above it. In z^* no group removes a resolved share: the spread is all observation error. In y^* the nuisance inputs together account for 28 % of the variance, and the lift curve (18 %) is the only single group above the resolution; the drag curve, air and mass, spin decay and knuckling each lie below it. The table removes one group at a time (fixed at mid-range, knuckling off, or $\sigma = 0.1$ mm). Its rows do not add, because the groups interact: the nuisance inputs matter mainly through how they map the observation error into the entry point, so removing the error removes their effect too. Better tracking, not better knowledge of the ball, would shrink the uncertainty most.

Table 8: Variance decomposition for the given kick: each row removes one group of sources. Share removed = 1−(s.d./s.d. of the full method)^2; rows do not add because the groups interact. Shares are shown only where they exceed the Monte Carlo resolution 15 % (Section 6.4); n.r.: not resolved, below it. The observation-error row rests on a small effective sample size and is indicative only.

Group removed	Parameters	s.d.		Share of variance removed	
		y^*	z^*	y^*	z^*
None (full method)	–	0.102 m	0.125 m	–	–
Lift curve	$C_{L,\max}$, Sp_L	0.0923 m	0.127 m	18 %	n.r.
Drag curve	$C_{D,\mathrm{sup}}$, $\Delta C_{D,\omega}$, Re_c	0.0969 m	0.126 m	n.r.	n.r.
Air and mass	ρ, m	0.0968 m	0.126 m	n.r.	n.r.
Spin decay	λ	0.0980 m	0.129 m	n.r.	n.r.
Knuckling	$\boldsymbol{\xi}$	0.0990 m	0.125 m	n.r.	n.r.
All nuisance inputs	$\boldsymbol{\phi}$, $\boldsymbol{\xi}$	0.0866 m	0.126 m	28 %	n.r.
Observation error	σ	0.00335 m	0.00280 m	100 %	100 %

Wind is not modelled (Assumption A7): a 2 m s^{-1} crosswind would move a trajectory with a given launch by 0.081 m (Table 5), but re-inferring the given kick with that wind known moves the answer by only 0.036 m (Section 9), because the observations already contain most of its effect on the early flight; unknown wind is part of the mismatch test.

6.5 Calibration

A 68 % region should contain the truth in 68 % of cases. For a test kick with true entry point $\mathbf{x}_{\text{true}}$ the *credibility level* η is the probability of the highest-density region whose boundary passes through $\mathbf{x}_{\text{true}}$,

$$\eta = P[\hat{p}(\mathbf{X}) > \hat{p}(\mathbf{x}_{\text{true}})]\,, \qquad \mathbf{X} \sim \hat{p}. \tag{18}$$

The truth lies inside R_α exactly when $\eta \le \alpha$; the *coverage* at level α is the share of test kicks with $\eta \le \alpha$. A calibrated predictor has coverage α at every level (η uniform on $[0,1]$); lower coverage means over-confidence. A stretch of every predictive cloud about its mode by a factor f,

$$\mathbf{x}_i^* \;\to\; \mathbf{x}_{\text{mode}} + f\,(\mathbf{x}_i^* - \mathbf{x}_{\text{mode}}), \tag{19}$$

multiplies the covariance by f^2 and leaves the shape and the mode unchanged; it can correct the overall width of the regions, not their shape (Section 7.3).

7 Testing the Method

On simulated kicks with known entry points, M4-B matches the exact spin fit M3 at the median (median error 0.17 m against 0.19 m; their paired-bootstrap difference, (0.023 ± 0.021) m, is about one standard deviation, Section 7.2), is better in the tail (90th percentile 0.39 m against 0.57 m) and never fails to predict, whereas M3 returns no crossing for 4 kicks. Its contribution is calibrated regions, the scoring probability and robustness under mismatch: the raw 68 % and 95 % regions contain the truth for 68 % and 90 % of in-model kicks, and for 68 % and 92 % in the configuration used for the given kick (Section 7.5). A conditional Gaussian mixture (Pedregosa et al., 2011) trained on the same simulator comes close at a fraction of the cost.

7.1 Synthetic Ground Truth

All predictors see the same simulated kicks, never used to train or tune them. Each has a launch speed uniform on $20\,\mathrm{m\,s^{-1}}$ to $38\,\mathrm{m\,s^{-1}}$ (a range that covers the mean speeds of (26 ± 3) and $(32 \pm 6)\,\mathrm{m\,s^{-1}}$ measured for club and World Cup penalties (Morya et al., 2005)), aimed uniformly at $|y| \le 4.5$ m, $0.3 \le z \le 3.2$ m (goals, frame hits, misses and ground contacts), spin, timing and nuisance parameters from their priors, and is flown with the full M4 including knuckling; its positions at t_1, t_2 get Gaussian errors of standard deviation σ. Kicks that touch the ground first count only in the scoring probability; 90 of the 120 in-model kicks reach the goal plane in flight. A predictor that returns no crossing for such a kick counts as a failure (infinite error).

The *in-model* set (120 kicks, $\sigma = 2.9$ mm) draws the truth from the model and priors the method assumes. The *mismatch* set (120 kicks) breaks them without telling the method: a drag plateau above its prior range, a lift curve partly outside, a larger spin scale, random wind, stronger knuckling, larger timing errors and rounded observations (Appendix B.5). Three *noise* sets (60 kicks each) use $\sigma = 5$, 10 and 20 mm, typical of ball tracking, and tell the method σ; the *rounding* set (120 kicks) rounds noise-free positions to 0.01 m, exactly as the given data, to test Assumption A4. We record the error $\epsilon = |\mathbf{x}_{\text{mode}} - \mathbf{x}_{\text{true}}|$, the credibility level η of the truth (Eq. (18)), whether the predicted ninth is right, and the runtime ($N = 1200$ for M4-B); for DM and the learned methods, coverage uses the Gaussian ellipse of the same probability. Here M4-B runs as the general method, with the timing offset free ($\sigma_\tau = 2$ ms, as in the test kicks); Section 7.5 repeats the in-model, rounding and mismatch tests for the exact kick instant used for the given kick.

Table 9: Accuracy on the in-model set, general method (timing offset free): distance between predicted point and truth (a kick with no predicted crossing counts as an infinite error), kicks with no crossing, coverage of the 68 % and 95 % regions (raw), share of the kicks entering the goal mouth whose predicted point lies in the true ninth (no crossing counts as wrong), and time per kick as in Table 6. ∞: no crossing for more than 10 % of the kicks.

Predictor	Median error	90th pct.	No crossing	68 % cov.	95 % cov.	Right ninth	Time
M0 straight line	0.87 m	1.5 m	0	–	–	12 %	3.0×10^{-4} ms
CA constant accel.	0.31 m	∞	11	–	–	62 %	3.2×10^{-4} ms
M2 drag only	0.45 m	0.76 m	5	–	–	61 %	1.3 ms
M3 exact spin fit	0.19 m	0.57 m	4	–	–	77 %	5.1 ms
DM delta method	as M3	as M3	as M3	59 %	88 %	as M3	7× M3
M4-B Bayesian	0.17 m	0.39 m	0	68 %	90 %	78 %	1700 ms
GRID3 grid lookup	1.0 m	5.4 m	–	68 %	96 %	48 %	0.0021 ms
COND conditioning	0.39 m	0.75 m	–	78 %	98 %	71 %	320 ms
CGMM mixture	0.18 m	0.37 m	–	66 %	93 %	87 %	90 ms

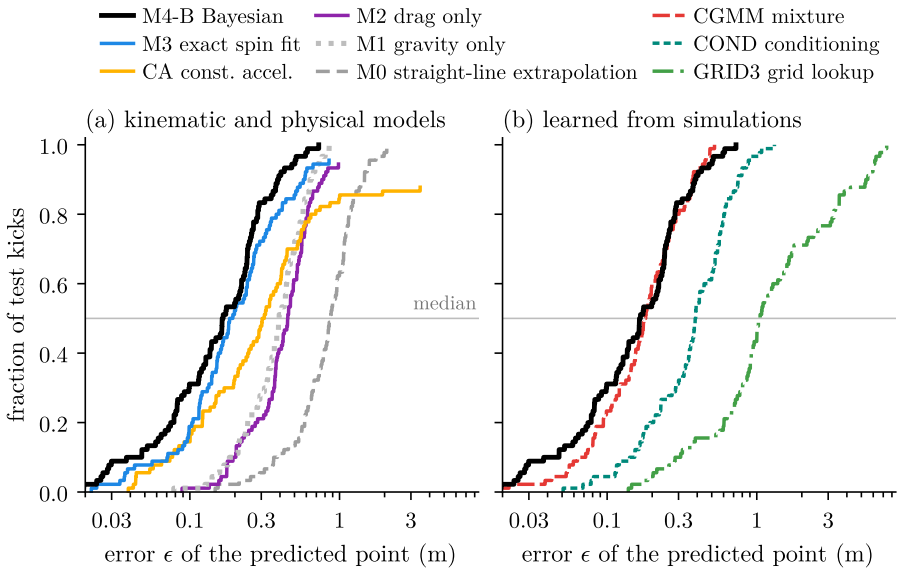


Figure 4: Error distribution on the in-model set: fraction of kicks predicted better than ϵ; curves further left are more accurate. Curves ending below 1: kicks with no predicted crossing count as failures. (a) Physical models: only those with the Magnus force, M3 and M4-B, reach the lowest errors; below them the order is not monotone in the physics, as drag without spin (M2) does worse than gravity alone (M1) and constant acceleration (CA) beats both. (b) Learned from simulations: only the mixture (CGMM) comes close to M4-B.

7.2 Accuracy

Only the predictors with a Magnus force reach the lowest errors (Table 9, Figure 4; M1 and GRID5 in Appendix B.5), and below them the order is not monotone in the physics. A straight line ignores gravity and the Magnus force, worth 0.73 m and 0.77 m for the given kick (Table 5), and misses by 0.87 m. CA reads the curl but freezes the early accelerations and finds no crossing for 11 kicks, yet at 0.31 m still beats both models without spin. Drag without spin (M2, 0.45 m) does even worse than gravity alone (M1, 0.39 m): neither has a force across the flight, although the data show one of $(11.1 \pm 2.5)\,\mathrm{m\,s^{-2}}$ (Section 4.4), and a fitted drag coefficient does not make up for it. M3 reaches 0.19 m and M4-B 0.17 m. To see whether this difference exceeds chance we use a paired bootstrap (Efron and Tibshirani, 1993): draw the test kicks 2000 times with replacement, compute both medians on each draw (a missing crossing counting as an infinite error) and take the standard deviation of their difference as the error of the observed one. The difference M3 − M4-B is (0.023 ± 0.021) m, about one standard deviation, so at the median the two are not distinguishable. M4-B gains in the tail (90th percentile 0.39 m against 0.57 m) and never fails, because its spin prior damps the implausibly large spins into which M3 turns rounding errors and its coefficients follow the Reynolds number as it falls along the flight (for the given kick from 4.5×10^5 to 3.6×10^5).

Among the learned predictors, the grid errs by 1.0 m, with near-nominal coverage only because its regions span whole ninths; dataset conditioning, short of nearby stored kicks (median effective sample size 39), by 0.39 m. The mixture nearly matches M4-B (0.18 m) at a fraction of its cost. It also puts more predictions in the right ninth (87 % against 78 %), a difference of a handful of kicks, comparable to the binomial noise of either share.

7.3 Calibration

In-model calibration is the direct test of the inference: the kicks come from the priors it assumes (up to the launch-velocity prior, which the data override), so wrong coverage would mean wrong inference or numerics, not wrong physics. The raw 68 % region contains the truth for 68 % of the 90 scored kicks, within one binomial standard error ($\sqrt{\alpha(1-\alpha)/n} = 4.9\,\%$) of nominal; the 95 % region for 90 %, about two standard errors (2.3 %) low (92 % in the configuration used for the given kick, Section 7.5). The whole curve is S-shaped (Figure 5): the 50 % and 68 % regions are right, while the predicted density is too broad in its core (small central regions contain the truth more often than their probability) and slightly too light in its tails (the 95 % region contains it less often). A single stretch (Eq. (19)) cannot fix this shape: fitted at 68 % on the even-numbered kicks it is $f^2 = 1.06$, essentially one (held-out coverage 73 % and 91 %), so we report raw regions, right at the 68 % level we headline and slightly optimistic at 95 %. An earlier version under-covered because one random knuckling realisation per sample in the window likelihood made the weights noisy; the knuckling covariance of Eq. (13) fixed it, raising the ratio of predictive standard deviations to a brute-force reference on eight in-model kicks (plain importance sampling of launch velocity, timing and spin) from 0.89 to 0.95. Grouped by the sign of the true spin, the mode falls short of the curl by (-0.074 ± 0.017) m; grouped by the predicted curl, the offset is (0.007 ± 0.017) m, consistent with zero: shrinkage towards zero spin, as in any posterior, not a bias given the data. On the rounding set the regions stay close to nominal (Table 10, Table 11), so replacing the rounding box by a Gaussian (Assumption A4) costs nothing measurable.

7.4 Noisier Data and Model Mismatch

The method degrades gradually (Table 10). Noise hurts M3 more than M4-B, whose spin prior limits how far noise can push the fitted spin; at 20 mm the error $\sqrt{5}\,\sigma/\Delta t^2$ of the measured acceleration (Section 4.4) exceeds the sideways acceleration of a typical curling kick, and the prediction leans on the priors. The mode then becomes unstable: it errs by 0.51 m, worse than

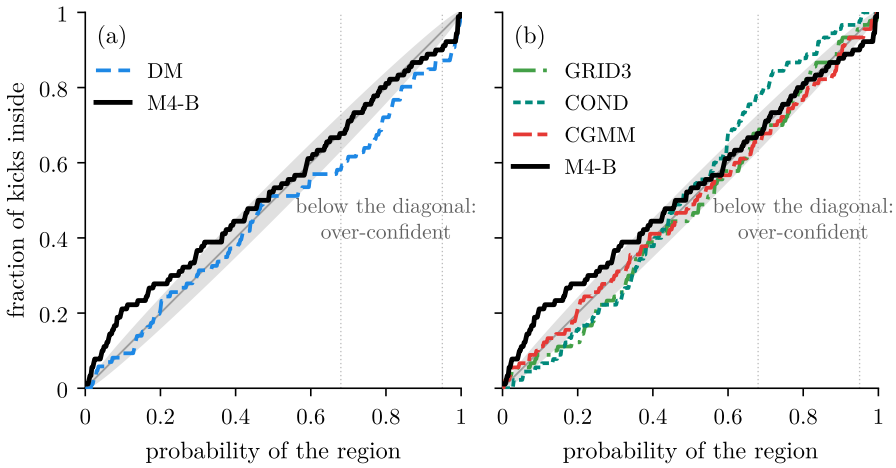


Figure 5: Calibration on the in-model set: fraction of kicks whose truth lies inside the region, against the probability of the region. Calibrated predictors lie on the diagonal (band: binomial standard error); below it, over-confident. (a) M4-B is S-shaped: its small central regions are too wide (21 % of the kicks inside the 10 % region, 37 % inside the 30 %), it is on the diagonal at 50 % and 68 % (52 %, 68 %) and slightly below it at 95 % (90 %): the predicted tails are slightly too light; the delta method is over-confident above 50 % (59 % at 68 %). (b) CGMM and GRID3 follow the diagonal; COND lies below it for small regions and above it for large ones: too narrow in the core, too wide in the tails.

Table 10: Robustness of the general method (timing offset free): median error of the M4-B mode and mean, of the exact spin fit M3, of gravity alone M1 (the baseline that ignores the curl) and of CGMM (no crossing counts as an infinite error), and raw coverage, on every test set (the learned methods were not run on the rounding set).

Test kicks	Median error					68 % coverage		95 % cov.
	M4-B mode	M4-B mean	M3	M1	CGMM	M4-B	CGMM	M4-B
In-model	0.17 m	0.16 m	0.19 m	0.39 m	0.18 m	68 %	66 %	90 %
Rounding only	0.18 m	0.17 m	0.23 m	0.42 m	–	64 %	–	97 %
$\sigma = 5$ mm	0.28 m	0.26 m	0.30 m	0.37 m	0.23 m	67 %	71 %	94 %
$\sigma = 10$ mm	0.38 m	0.38 m	0.57 m	0.42 m	0.37 m	61 %	59 %	93 %
$\sigma = 20$ mm	0.51 m	0.37 m	1.4 m	0.42 m	0.40 m	80 %	71 %	98 %
Mismatch	0.27 m	0.26 m	0.32 m	0.55 m	0.28 m	24 %	32 %	60 %

gravity alone (M1, 0.42 m), the posterior mean by 0.37 m. Sampling is not the cause (median effective sample size 220 of $N = 1200$; kicks with more do no better, and a kernel twice as wide brings the mode to the mean's error): the density is broad and skewed, and the median distance between mode and mean grows from 0.048 m in-model to 0.21 m. The mixture, also told σ, matches or beats the M4-B mode on the noise sets.

Under mismatch the method explains the observed accelerations with a wrong ball, extrapolates that to the goal, and cannot tell wind from a change of launch direction (Assumption A7): its median error grows to 0.27 m (M3: 0.32 m) and its 68 % coverage falls to 24 %, a weakness of the method. The learned methods inherit the simulator and under-cover too: 32 % for the mixture, 38 % for conditioning. If the true physics lies as far outside our priors as in this test, the regions must be widened by $f = 1.7$ (Eq. (19), fitted so that the 68 % regions cover 68 % of the mismatch kicks, which brings the 95 % coverage to 94 %) to stay honest. Kicks from our own simulator cannot reveal an error it shares with the predictors built on it, and the mismatch set probes only the departures we built into it.

7.5 The Configuration Used for the Given Kick

The given kick is predicted with the kick instant exactly as stated ($\sigma_\tau = 0$). We test this configuration on kicks whose true instant is exact: the in-model, rounding and mismatch kicks of Section 7.1 with $\tau = 0$ (Table 11). Its regions are close to nominal on in-model and rounding kicks. Its mode matches M3 at the median (paired-bootstrap difference (0.000 ± 0.021) m, as in Section 7.2) and the posterior mean is slightly better; under mismatch it errs less than M3 and never fails, while M3 finds no crossing for 5 kicks. Under mismatch its regions under-cover; widening them by $f = 1.64$ (Eq. (19), fitted so that the 68 % regions cover 68 % of these kicks) brings the 95 % coverage to 91 %, and the factor fitted on the even-numbered kicks alone gives 66 % and 92 % on the odd ones. For the given kick it widens the 68 % half-widths to 0.25 m in y^* and 0.33 m in z^* (Section 8.1).

Table 11: The configuration used for the given kick (exact kick instant, $\sigma_\tau = 0$) on test kicks with exact timing: median error of the M4-B mode and mean and of M3 (no crossing counts as an infinite error; M3's kicks without a crossing in brackets), and raw coverage of the M4-B regions.

Test kicks (exact timing)	Scored	M4-B mode	M4-B mean	M3	68 % cov.	95 % cov.
In-model	90	0.18 m	0.17 m	0.18 m (3)	68 %	92 %
Rounding only	75	0.19 m	0.17 m	0.19 m (1)	63 %	96 %
Mismatch	85	0.24 m	0.25 m	0.27 m (5)	40 %	67 %

7.6 Failure Modes

The method fails in five identifiable situations, still returning a density, but one that is too narrow, answers a question we excluded, or has a misleading peak. *Ground contact:* a low shot is predicted as "does not reach the goal plane" (Assumption A3); the clean-goal probability has Brier score (mean squared difference from the 0/1 outcome) 0.051. *Posts and bar:* an error of 0.17 m cannot say on which side of a post the ball ends up; the method gives the probability of striking the frame. *Knuckleballs:* knuckling moves a non-spinning ball randomly by about 0.087 m over the flight, and the push already received in the observation window is not carried on to the goal (Appendix B.4); at the given precision the tests do not resolve this, as kicks with $\mathrm{Sp}_0 < \mathrm{Sp}_K$ have the same median error, 0.19 m, as the third with most spin (0.19 m). *Wind:* two observations cannot tell a steady crosswind from a slightly different launch direction; a known wind enters M4 through $\mathbf{v}_{\mathrm{rel}}$. *Very noisy data:* the mode becomes unstable (Section 7.4). The problem asks for the most likely point, so our headline is the mode, nearly as accurate as the

mean at the given precision (0.17 m against 0.16 m); for much noisier data, report the posterior mean.

8 Application to the Given Scenario

The given kick is a hard, strongly curling shot high to the goalkeeper's right that enters the goal inside the frame, too late for any dive that starts after the second observation.

8.1 The Predictive Density

The answer to task (T4) is:

> The ball most likely enters the goal high on the goalkeeper's right, at $y^* = (2.67 \pm 0.10)$ m, $z^* = (1.97 \pm 0.13)$ m (± 1 s.d.; total with the spin-prior systematic: ± 0.11 m in y, ± 0.13 m in z; 68 % region ± 0.15 m in y and ± 0.20 m in z, 95 % region ± 0.25 m in y and ± 0.35 m in z); it enters cleanly with probability 0.999 (0.960 allowing for model error); it crosses the goal line (0.392 ± 0.003) s after the kick, leaving the goalkeeper 0.29 s after the second observation, less than the time an instantaneous dive needs to reach it and far less than the 0.6 s measured for real dives.

The $\pm$ values are standard deviations of the predictive density (Section 6) and the regions its highest-density regions, as computed, without recalibration; the totals add the spin-prior systematic of Section 9 in quadrature. On test kicks with the kick instant exact, as here, the 68 % and 95 % regions contain the truth in 68 % and 92 % of kicks (Section 7.5): they are honest if the ball and the air lie within our priors. If the true physics lies as far outside them as in the mismatch test, where the 68 % regions cover only 40 % of kicks, the regions must be widened by the factor 1.64 to stay honest (Table 12; after widening, the mismatch 95 % regions cover 91 %). The most likely point stays; the probability of a clean goal falls to 0.960 and that of striking the frame rises to 0.039.

Table 12: Half-widths of the uncertainty regions of the given kick, as computed (in-model, the headline) and widened for model error by the factor fitted on the mismatch test with exact kick instants.

Regions	68 %, y^*	68 %, z^*	95 %, y^*	95 %, z^*
In-model (headline)	± 0.15 m	± 0.20 m	± 0.25 m	± 0.35 m
With model error ($\times 1.64$)	± 0.25 m	± 0.33 m	± 0.41 m	± 0.57 m

Figure 6 shows the density on the goal face: the 68 % region is a compact oval high on the goalkeeper's right, nearer the post than the centre, and the 95 % region lies entirely inside the clean-entry rectangle, clear of post and bar. The mean (2.66 m, 1.97 m) lies close to the mode (standard deviations 0.10 m and 0.13 m, correlation 0.1). A simulated ball reaches the goal plane with probability 1.00, strikes the frame with probability below 10^{-3} and practically never passes outside it.

8.2 All Methods on This Kick

M3, the other physical model with the Magnus force, and the mixture (CGMM) agree with the headline to well within one standard deviation; the physical models without the Magnus force fall outside the 95 % region (Table 13).

The disagreements follow the physics. M0 ignores gravity and goes over the bar; M1 and M2 cannot curl the ball and place it too far towards the right post; the constant-acceleration model holds the deceleration of t_1 for the whole flight, so the ball arrives late and curls too

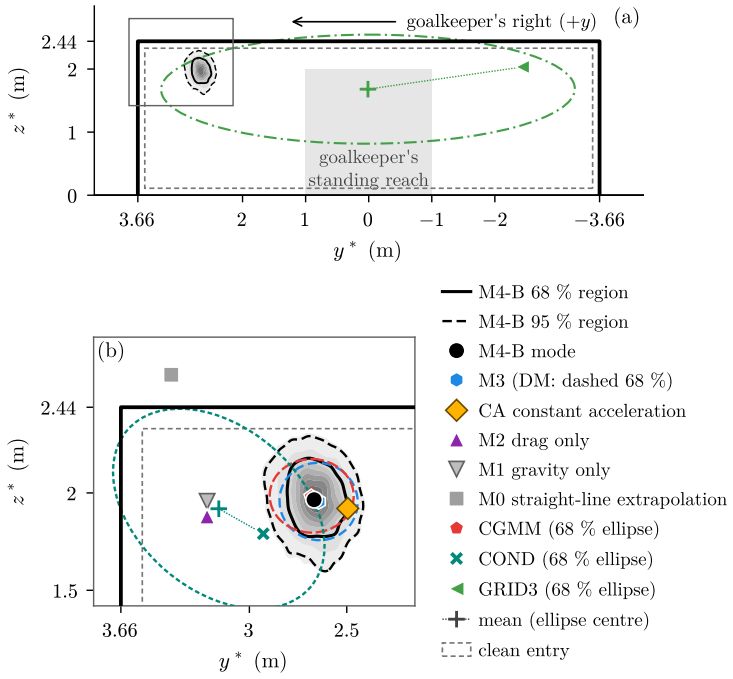


Figure 6: The ball goes high on the goalkeeper's right, well inside the frame; models without the Magnus force (M0–M2) fall outside the 95 % region, and the grid lookup on the wrong side of the goal. The goal face is seen from the penalty mark, so $+y$, the goalkeeper's right, points to the left. (a) The whole goal, with the goalkeeper's standing reach (shaded), the region of the enlargement (box) and the grid prediction GRID3 (green) with its 68 % ellipse. (b) Enlargement: the predictive density (grey), its 68 % (solid) and 95 % (dashed) regions and the M4-B mode (black); M0 (grey square, straight-line extrapolation of the launch velocity), M1 (grey triangle), CA (orange), M2 (purple) and M3 (blue); the 68 % ellipses of DM (blue dashed, centred on M3), CGMM (red) and COND (teal); and the clean-entry rectangle (thin dashed). Markers show each method's mode; for GRID3, COND and CGMM a cross marks the mean, the centre of the ellipse, joined to the mode by a dotted line.

Table 13: Predictions for the given kick, their distance from the M4-B mode and, for methods that return a density, P_{clean}. DM is the M3 fit with spread (0.13 m, 0.13 m).

Code	Method	y^*	z^*	Distance	P_{clean}
M0	straight-line extrapolation	3.40 m	2.61 m	0.97 m	–
M1	gravity only	3.22 m	1.96 m	0.55 m	–
CA	constant acceleration	2.49 m	1.92 m	0.18 m	–
M2	drag only	3.22 m	1.88 m	0.56 m	–
M3, DM	exact spin fit	2.64 m	1.96 m	0.025 m	–
GRID3	grid lookup (5×5 identical)	(-2.4 ± 2.2) m	(2.0 ± 0.6) m	5.1 m	0.34
COND	dataset conditioning	(2.9 ± 0.4) m	(1.8 ± 0.3) m	0.31 m	0.72
CGMM	conditional Gaussian mixture	(2.68 ± 0.14) m	(1.98 ± 0.13) m	0.021 m	0.99
M4-B	Bayesian (headline)	(2.67 ± 0.10) m	(1.97 ± 0.13) m	–	0.999

much, landing between the 68 % and the 95 % region. M3 lands close to the headline. The learned predictors are the fast, deployable version of M4-B: trained on the same simulator, they can at best approach the Bayesian answer. The mixture (CGMM) essentially reproduces it, in 90 ms instead of 17 s for M4-B with 20000 samples; dataset conditioning is wider and lies off the headline because few stored kicks resemble this one (effective sample size 34). The grid lookup shows why the physics is needed: the ball climbs above the line from the mark to the crossbar so early that both observations lie outside every pyramid, so the lookup can only return what all such kicks do. Most of them do not enter the goal, the three top ninths are almost equally likely, and the one it picks lies on the wrong side of the goal.

8.3 The Kick Itself

The posterior describes a hard, curled shot: launch speed $(32.35 \pm 0.07)\,\mathrm{m\,s^{-1}}$, at $(12.26 \pm 0.21)^\circ$ above the horizontal and $(17.16 \pm 0.17)^\circ$ towards the goalkeeper's right, within the $(32 \pm 6)\,\mathrm{m\,s^{-1}}$ measured for World Cup penalties (Morya et al., 2005); perpendicular spin $(45 \pm 13)\,\mathrm{rad\,s^{-1}}$, with its axis $(17 \pm 7)^\circ$ from the downward vertical (mostly sidespin, a little backspin); spin parameter at launch $\mathrm{Sp} = 0.15 \pm 0.04$; lift coefficient $C_L = 0.20 \pm 0.04$, below the $C_L \approx 0.35$ reached at $\mathrm{Sp} \approx 0.5$ (Goff and Carré, 2012), as expected at this lower spin parameter. Spin and axis agree with the model-free reading of Section 4.4 within their spreads. The launch speed lies close to the finite-difference value $32.6\,\mathrm{m\,s^{-1}}$, which assumes a constant acceleration over the first 0.1 s; with the launch instant free it would be $(32.01 \pm 0.16)\,\mathrm{m\,s^{-1}}$, and the crossing time (0.390 ± 0.003) s instead of (0.392 ± 0.003) s (Table 4). With the instant fixed, the data pull the drag and the air density up within their priors (Section 4.4) and leave the lift curve, the critical Reynolds number and the spin decay at their priors.

8.4 Can the Goalkeeper Save It?

No goalkeeper who waits for the second observation can reach this ball. We model him standing on the goal line at $y = 0$, his hands covering $|y| \le 1.0\,\mathrm{m}$, $z \le 2.0\,\mathrm{m}$; to reach a point (y, z) outside this envelope he must move them

$$d(y,z) = \sqrt{\max\big(|y| - 1.0,\,0\big)^2 + \max\big(z - 2.0,\,0\big)^2}\ \mathrm{m}. \tag{20}$$

The algorithm takes three steps: (1) the goalkeeper starts a reaction time t_r after the second observation; (2) he moves his hands at a constant dive speed v_d, covering $v_\mathrm{d}(t^* - t_2 - t_r)$ by the crossing time t^*; (3) a simulated ball counts as saved if this covers $d(y^*, z^*)$, and the probability of a save is the posterior weight of the saved samples. We take $v_\mathrm{d} = 3,\ 4,\ 5\,\mathrm{m\,s^{-1}}$ and $t_r = 0.15,\ 0.20,\ 0.30\,\mathrm{s}$, assumptions of this section only.

On average the goalkeeper must move his hands 1.7 m; even an instantaneous dive at $5\,\mathrm{m\,s^{-1}}$ needs 0.33 s, more than the 0.29 s that remain. The probability of a save (Fig. 7) is zero at every dive speed and reaction time of the grid: 0.0 for a realistic goalkeeper ($v_\mathrm{d} = 4\,\mathrm{m\,s^{-1}}$, $t_r = 0.20\,\mathrm{s}$) and 0.0 even in the best case ($v_\mathrm{d} = 5\,\mathrm{m\,s^{-1}}$, $t_r = 0.15\,\mathrm{s}$). Told the answer at t_2 and moving at once at $5\,\mathrm{m\,s^{-1}}$, he saves with probability 0.018. A save probability of one half would need a dive speed of $5.7\,\mathrm{m\,s^{-1}}$ or, at $5\,\mathrm{m\,s^{-1}}$, a reaction time of $-0.041\,\mathrm{s}$; reaction times are counted from t_2, so this negative value means starting before the second observation. The limit is the time left, not the prediction: to save such a kick a goalkeeper must commit before t_2, from the kicker's run-up and body (van der Kamp et al., 2018), or from the first hundredths of a second of flight, where our method would apply with a larger observation error.

9 Sensitivity Analysis

In every variant the most likely entry point stays inside the frame. Only the width of the spin prior moves it by about one s.d. of y^* or more, and only in y^*; the drag prior moves it by less

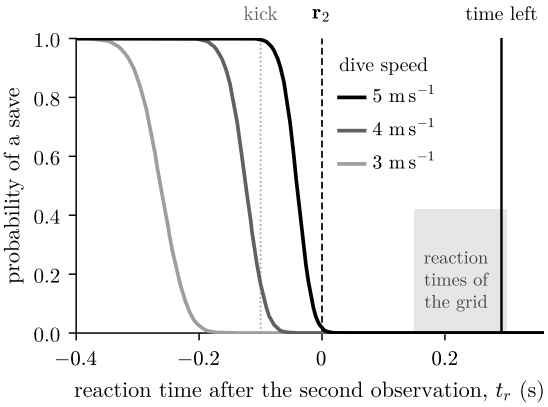


Figure 7: The prediction comes too late to act on: even a goalkeeper who knows the answer at the second observation and moves at once almost never reaches the ball. Probability of a save against the reaction time t_r, counted from the second observation, for dive speeds of 3, 4 and 5 m s^{-1} (light to dark). Vertical lines: the kick ($t_r = -t_2$, dotted), the second observation ($t_r = 0$, dashed) and the time left until the ball crosses the goal line (solid); the shaded band marks the reaction times of the grid. A save becomes likely only for negative t_r, a start before the second observation.

than half a combined s.d., and every other assumption by no more than about twice the Monte Carlo noise. The spin prior is therefore the assumption a reader should scrutinise.

We repeat the inference of Section 5 for the given kick with one assumption changed at a time: spin prior $\sigma_\omega = 10$ or 50 rad s^{-1} instead of 25 (Assumption A13); air density and ball mass fixed at 1.2 kg m^{-3} and 0.43 kg (Assumptions A1 and A8); a wide drag prior, $C_{D,\mathrm{sup}}$ between 0.10 and 0.30 and $\Delta C_{D,\omega}$ between 0 and 0.25 (Assumption A10); no knuckling (Assumption A12); a known crosswind of 2 m s^{-1} towards $+y$ (Assumption A7); the launch instant free with $\sigma_\tau = 2$ ms (Assumption A2); and larger observation errors (Assumption A4). Each shift is measured from a reference run of the unchanged method with the same random seed and sample size (8000 samples for the five prior variants, 20000 for the crosswind and the launch instant), so that most of the Monte Carlo noise cancels. Repeating both reference runs with four seeds, their modes scatter by 0.028 m and 0.021 m; a shift of at most about twice this noise is not resolved (Table 14, Fig. 8).

Table 14: Sensitivity of the prediction for the given kick. "Shift" is the distance of the mode from the reference run above it, also given in units of the combined standard deviation $(s_y^2 + s_z^2)^{1/2}$ of the reference and of the Monte Carlo noise of the mode at the same sample size. The smaller reference run differs from the headline only by Monte Carlo noise. The last rows repeat the headline with larger observation errors (s.d. only).

Variant	Assumption	y^* mode	z^* mode	s.d. y^*	s.d. z^*	Shift	Shift/s.d.	Shift/noise
Reference run	–	2.65 m	1.98 m	–	–	–	–	–
$\sigma_\omega = 10$ rad s^{-1}	A13	2.83 m	1.98 m	0.10 m	0.11 m	0.18 m	1.1	6.3
$\sigma_\omega = 50$ rad s^{-1}	A13	2.54 m	2.00 m	0.10 m	0.14 m	0.11 m	0.64	3.7
Wide drag prior	A10	2.70 m	1.93 m	0.11 m	0.12 m	0.072 m	0.44	2.5
Fixed air and mass	A1, A8	2.65 m	1.97 m	0.099 m	0.13 m	0.017 m	0.10	0.58
No knuckling	A12	2.65 m	1.97 m	0.10 m	0.13 m	0.010 m	0.061	0.35
Headline run	–	2.67 m	1.97 m	0.10 m	0.13 m	–	–	–
Crosswind 2 m s^{-1}	A7	2.70 m	1.97 m	0.096 m	0.13 m	0.036 m	0.22	1.7
Launch instant free	A2	2.70 m	1.96 m	0.11 m	0.12 m	0.029 m	0.17	1.3
$\sigma = 5$ mm	A4	–	–	0.17 m	0.22 m	–	–	–
$\sigma = 10$ mm	A4	–	–	0.24 m	0.27 m	–	–	–
$\sigma = 20$ mm	A4	–	–	0.33 m	0.34 m	–	–	–

The spin prior. A narrow prior, $\sigma_\omega = 10$ rad s^{-1}, pulls the inferred spin towards zero; with less Magnus force the ball curls less back towards the centre, and the mode shifts by 0.18 m in y^*, towards the goalkeeper's right (1.8 s.d. in y^*, 1.1 combined), while z^* hardly moves. The inferred perpendicular spin, (45 ± 13) rad s^{-1}, lies more than four standard deviations out in

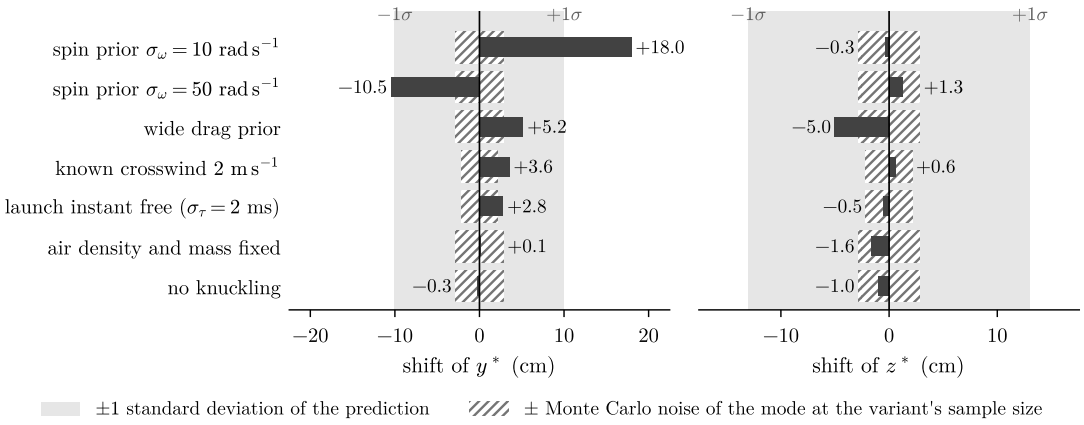


Figure 8: Only the spin prior moves the answer by about one standard deviation or more, and only in y^*: the narrow prior, which the data disfavour, by 0.18 m, the wide prior by −0.10 m; every other assumption moves it by at most 0.072 m (the wide drag prior, 2.5 times the Monte Carlo noise). Shift of the most likely entry point of the given kick in y^* (left) and z^* (right) when one assumption is changed (variant mode minus the mode of its reference run with the same seed and sample size; crosswind and launch instant free: the headline run), sorted by size. Grey band: ±1 standard deviation of the predictive density; hatched band: ±Monte Carlo noise of the mode at the variant's sample size (spread of the mode over four seeds; the combined y–z value is drawn on each axis, so the band is conservative).

that prior, and the data disfavour it against $\sigma_\omega = 25$ by a Bayes factor of 16. Between 25 and 50 rad s^{-1} the data barely discriminate: a factor of 1.8 in favour of 50, too small to count as evidence. The wide prior shifts the mode towards more curl, by −0.10 m in y^* (−1.0 s.d. in y^*, 0.64 combined). We keep 25, the width on which the method is validated, and add the choice as a systematic: weighting the 25 and 50 posteriors by their evidence (weight 0.64 on 50) shifts the posterior mean by −0.049 m in y^* and 0.0066 m in z^*. In quadrature with the predictive spread this gives the total standard deviations of the answer, 0.11 m in y^* and 0.13 m in z^* (against 0.10 m and 0.13 m without it).

The remaining assumptions. With the kick instant fixed, the data pull the drag into the upper part of its prior (Section 4.4); a wider drag prior lets it go further and moves the mode by 0.072 m, resolved but 0.44 of a standard deviation. Freeing the launch instant instead trades the drag against an earlier launch and leaves the entry point (Section 4.4). The observed accelerations fix κC_D and κC_L, not κ, so fixing air density and mass moves the inferred coefficients, not the trajectory. Knuckling is switched off by the spin (factor 0.05 ± 0.04). A 2 m s^{-1} crosswind displaces a trajectory with a given launch by about 0.10 m (simulated: 0.081 m; Table 5), but known and added to the model it moves the re-inferred mode by only 0.036 m: the observations already contain most of the wind's effect on the early flight, and the inference attributes it to a slightly different launch and spin (perpendicular spin (48 ± 14) rad s^{-1} against (45 ± 13) rad s^{-1}). These four shifts are not resolved. Larger observation errors matter most: at $\sigma = 5$, 10 and 20 mm the standard deviation in y^* grows to 0.17 m, 0.24 m and 0.33 m, and the probability of a clean goal falls from 0.999 to 0.92, 0.91 and 0.71, so Assumption A4 carries the size of the answer's uncertainty. In every variant the conclusion of Section 8 — high on the goalkeeper's right, inside the frame, too late to save — holds.

10 Strengths and Weaknesses

The method's main strength is that every physical coefficient is anchored either in published measurements or in the data themselves, and its regions are calibrated; its main weakness is that it has been tested only against its own simulator, and its regions are honest only when the ball and the air match its assumptions.

10.1 Strengths

Coefficients anchored in measurements and in the data.
Every parameter range of Table 2 comes from wind-tunnel or trajectory measurements of real footballs (Hong and Asai, 2014; Goff and Carré, 2009, 2012) or is labelled as our assumption and varied in Section 9. The data then select within those ranges: the sideways acceleration is more than four standard errors and 4.7 times what knuckling could produce, so the presence of spin is measured; its size depends on the lift curve (30–86 $\mathrm{rad\,s^{-1}}$, 68 %).

A verified integrator.
The exact fit of the integrated model M3 reproduces the coefficients read by finite differences ($C_L = 0.211$ against 0.21 ± 0.05, $C_D = 0.378$ against 0.38 ± 0.06), and a step eight times smaller moves the entry point by less than 10^{-12} m (Appendix A.1).

Calibrated regions, never failing, robust under mismatch.
On test kicks with the kick instant exact, as for the given kick, the 68 % and 95 % regions contain the truth in 68 % and 92 % of kicks (Section 7.5); the method always returns a prediction, while the exact spin fit fails on 3 of 90. Its median error, 0.18 m, matches the exact spin fit's (0.18 m), and its posterior mean is slightly better (0.17 m); under deliberately broken assumptions (drag and lift outside the prior, wind, stronger knuckling) it is better (0.24 m against 0.27 m), and with 20 mm observation errors the exact spin fit fails (1.4 m) while the posterior mean stays at 0.37 m (Section 7.4).

Two comparable uncertainty routes.
The delta method linearises M3 and propagates only the rounding error, without any prior: standard deviations (0.13 m, 0.13 m), comparable to the Bayesian (0.10 m, 0.13 m), although its 68 % ellipses cover only 59 % of in-model kicks. Without the rounding almost nothing of the Bayesian spread remains (100 % of the y^* variance vanishes), while fixing every property of ball and air removes 28 % of the y^* variance and essentially none of z^* (Section 6.4).

General, and fast where it must be.
Nothing in the method refers to the given kick; it accepts any pair of observations for launch speeds of 15 $\mathrm{m\,s^{-1}}$ to 40 $\mathrm{m\,s^{-1}}$ and a two-sided kick-timing uncertainty for tracking data. The full density takes 17 s with 20000 samples; the conditional Gaussian mixture, trained on the same simulator, predicts $((2.68 \pm 0.14)\,\mathrm{m}, (1.98 \pm 0.13)\,\mathrm{m})$ for the given kick in 90 ms, fast enough to run within the flight.

It answers what a goalkeeper needs.
Besides a point, the method returns the probabilities of a clean goal, a frame hit and a miss, and the crossing time (Section 8.4).

10.2 Weaknesses

Over-confident when the assumptions fail.
On kicks that break our assumptions the 68 % region contains the truth in only 40 % of kicks. Widening every region by the factor 1.64 restores honest coverage there (95 %: 91 %) and gives the model-error regions of Table 12, but the factor is empirical and fitted on one simulated mismatch set.

Rounding as the only observation error.
The size of the uncertainty rests on Assumption A4: with 20 mm errors the standard deviation in y^* grows to 0.33 m and the probability of a clean goal falls to 0.71 (Section 9).

Still air.
Two observations cannot separate a steady crosswind from a slightly different launch direction. For the given kick a known $2\,\mathrm{m\,s^{-1}}$ crosswind moves the re-inferred answer by only 0.036 m, but an unknown wind is part of what the mismatch test penalises.

Drag not identified.
Two observations cannot tell more drag from a kick a fraction of a millisecond earlier; with the instant exact the drag sits in the upper part of its prior, and the answer does not depend on the choice (Section 4.4).

The spin prior.
The data disfavour a narrow spin prior but barely discriminate between $\sigma_\omega = 25$ and $50\,\mathrm{rad\,s^{-1}}$. Weighting the two by their evidence shifts the prediction by -0.049 m in y^* and 0.0066 m in z^*, a systematic that the regions do not include but the answer's total standard deviations (0.11 m, 0.13 m) do (Section 9).

Knuckling during the observations.
A ball with little spin is already moved 0.0086 m by knuckling at the second observation, more than the 2.9 mm rounding, and 0.087 m at the goal. Adding this variance to the observation variance makes such kicks honestly less predictable, not more accurate; the given kick spins too fast for it to matter.

A finite test set.
Coverage is measured on the 90 in-model kicks that reach the goal plane, with a binomial standard error of 4.9 % on the 68 % coverage, and the median errors of the best predictors differ by about their own sampling error. No test compares a prediction with a real tracked ball.

The mode of a skewed density.
We report the most likely point, as asked; the mean (2.66 m, 1.97 m) minimises the expected squared error and lies close to it here, but is the better report for very noisy data (Section 7.4).

A rigid ball without panel orientation.
Hong and Asai (2014) found that the orientation of the panels relative to the flow changes the drag and the fluctuating side forces; we average over ball designs but not over orientations within one flight.

11 Conclusion

Two positions of the ball in the first tenth of a second are enough to say where a penalty kick will enter the goal and how sure that is, but not soon enough for a goalkeeper to act on it.

We predicted the entry point with a physical flight model — gravity, drag with the drag crisis, the Magnus force, spin decay and knuckling — whose coefficients come from wind-tunnel and trajectory measurements, and inferred its unknowns by Bayesian importance sampling: plausible kicks are weighted by how well they reproduce the observations and flown to the goal.

The data speak clearly before any model is fitted. The ball accelerates sideways at $(11.1 \pm 2.5)\,\mathrm{m\,s^{-2}}$, more than four standard errors and 4.7 times what knuckling could produce: the kick is a hard shot curling back towards the centre of the goal. Its drag, by contrast, the two observations fix only together with the exact instant of the kick, and the answer does not depend on how that trade-off is resolved (Section 4.4). For the given kick, the answer is:

> The ball most likely enters the goal high on the goalkeeper's right, at $y^* = (2.67 \pm 0.10)$ m, $z^* = (1.97 \pm 0.13)$ m (± 1 s.d.; total with the spin-prior systematic: ± 0.11 m in y, ± 0.13 m in z; 68 % region ± 0.15 m in y and ± 0.20 m in z, 95 % region ± 0.25 m in y and ± 0.35 m in z); it enters cleanly with probability 0.999 (0.960 allowing for model error); it crosses the goal line (0.392 ± 0.003) s after the kick, leaving the goalkeeper 0.29 s after the second observation, less than the time an instantaneous dive needs to reach it and far less than the 0.6 s measured for real dives.

On simulated kicks with the kick instant exact, the method's median error, 0.18 m, matches that of an exact spin fit (0.18 m), but unlike the fit it never fails, and its 68 % and 95 % regions contain the truth in 68 % and 92 % of kicks (Section 7.5); if the ball or the air lie as far outside our priors as in the mismatch test, the regions must be widened by 1.64 (Table 12). In every variant we tried the answer stays inside the frame; the width of the spin prior, whose effect enters the total standard deviations of the answer, is the assumption that deserves scrutiny. Even a goalkeeper told the answer at the second observation and moving at once would save the ball with probability only 0.018: the limit is time, not prediction.

The improvement that would matter most is real tracking data: a few hundred tracked kicks with measured entry points would test the method outside its simulator, pin down the drag and lift curves, separate drag from the kick instant, and measure the calibration on real balls.

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A Details of the Flight Model

A.1 Numerical Integration

Equation (1) has no closed-form solution for M2–M4, so we integrate it with the classical fourth-order Runge–Kutta method (Press et al., 2007), with a step of 2 ms for the flight to the goal and 2.5 ms up to the second observation. The algorithm advances each trajectory step by step and, after every step, checks three events. First, if an observation time t_1 or t_2 lies inside the step, the position at that time is found by cubic Hermite interpolation between the step's end points (Press et al., 2007, Sec. 17.2.2); this uses both positions and velocities and is therefore accurate to the same order as the integrator. Second, if x passes L inside the step, the crossing

time is found by Newton's method (Press et al., 2007, Sec. 9.4) on the interpolated $x(t)$, and the crossing point (y^*, z^*) and time t^* are recorded. Third, the trajectory ends if its centre falls below $z = r_b$ (ground contact, Assumption A3) or if $v_x \le 0$. The knuckling process $\boldsymbol{\xi}$ is updated once per step, which is accurate because the step is much shorter than its correlation time τ_c. The integrator advances many trajectories at once as arrays, using NumPy (Harris et al., 2020). Repeating the given kick with a step eight times smaller changes the entry point by less than $10^{-12}\,\mathrm{m}$, far below the precision of the data.

A.2 Calibration of the Knuckling Force

The knuckling acceleration of Eq. (8) has one amplitude, a_{30}, its size at $30\,\mathrm{m\,s^{-1}}$. Hong and Asai (2014) kicked non-spinning balls with a robot and measured impact points scattered with standard deviations of 0.13 m to 0.68 m at 25 m. We calibrate in three steps.

1. Simulate 6000 non-spinning kicks with model M4 at $30\,\mathrm{m\,s^{-1}}$ and 15° elevation, with nominal air and ball and $a_{30} = 1\,\mathrm{m\,s^{-2}}$, and measure the standard deviation of their crossing points in the plane $x = 25\,\mathrm{m}$.

2. Because the displacement caused by a small random force is linear in its amplitude, scale a_{30} so that this standard deviation becomes 0.30 m, the middle of the measured range.

3. Check the result with a new set of random realisations.

The result is $a_{30} = 2.15\,\mathrm{m\,s^{-2}}$, which reproduces a scatter of 0.30 m and corresponds to a force with a standard deviation of 0.93 N per perpendicular component, within the standard deviations of 0.6 N to 3.1 N measured for the side and lift forces in the wind tunnel. Over the 11 m of a penalty, the same force scatters a non-spinning ball by 0.087 m.

The correlation time $\tau_c = 0.06\,\mathrm{s}$ is our assumption. It corresponds to force fluctuations of a few hertz; Hong and Asai (2014) found the spectral amplitude of the fluctuating force to rise near 2.5 Hz. The suppression scale $\mathrm{Sp}_K = 0.05$ is also our assumption (Assumption A12); for the given kick it reduces the force to 0.05 ± 0.04 of its non-spinning value, so its value does not matter (Section 9).

B Inference Details

This appendix gives the details of the algorithm of Section 5.3 that the main text states in one sentence each.

B.1 The Spin Prior in Terms of the Lift Vector

The lift vector $\boldsymbol{\ell}$ of Eq. (11) has two components along the unit vectors $\hat{\mathbf{e}}_1 = \hat{\mathbf{e}} \times \hat{\mathbf{z}} / |\hat{\mathbf{e}} \times \hat{\mathbf{z}}|$ (horizontal) and $\hat{\mathbf{e}}_2 = \hat{\mathbf{e}} \times \hat{\mathbf{e}}_1$ (nearly vertical). The map back to the spin keeps the direction, rotated by a right angle, and inverts Eq. (6) for the size:

$$\omega_\perp(\boldsymbol{\ell}) = \frac{v_0}{r_b}\,\mathrm{Sp}_0, \qquad \mathrm{Sp}_0 = -\mathrm{Sp}_L \ln\!\left(1 - \frac{|\boldsymbol{\ell}|}{C_{L,\max}}\right), \qquad \boldsymbol{\omega}_0 = \omega_\perp \frac{\hat{\mathbf{v}}_0 \times \boldsymbol{\ell}}{|\hat{\mathbf{v}}_0 \times \boldsymbol{\ell}|} + \omega_\parallel\, \hat{\mathbf{e}}, \tag{21}$$

which requires $|\boldsymbol{\ell}| < C_{L,\max}$: no spin produces more lift than the plateau.

The change of variables must carry the spin prior over exactly, or the method would silently assume a different prior. For a polar map that keeps the angle and sends the radius $|\boldsymbol{\ell}|$ to $\omega_\perp(|\boldsymbol{\ell}|)$, the Jacobian determinant is $(\omega_\perp/|\boldsymbol{\ell}|)\,(\mathrm{d}\omega_\perp/\mathrm{d}|\boldsymbol{\ell}|)$, so

$$p(\boldsymbol{\ell} \mid \mathbf{v}_0, \boldsymbol{\phi}) = \mathcal{N}_2\big(\boldsymbol{\omega}_\perp(\boldsymbol{\ell}); \mathbf{0}, \sigma_\omega^2 \mathbf{I}\big)\, \frac{\omega_\perp}{|\boldsymbol{\ell}|}\, \frac{\mathrm{d}\omega_\perp}{\mathrm{d}|\boldsymbol{\ell}|}, \qquad \frac{\mathrm{d}\omega_\perp}{\mathrm{d}|\boldsymbol{\ell}|} = \frac{v_0}{r_b}\, \frac{\mathrm{Sp}_L}{C_{L,\max} - |\boldsymbol{\ell}|}. \tag{22}$$

As $|\boldsymbol{\ell}| \to 0$ the ratio $\omega_\perp/|\boldsymbol{\ell}|$ tends to the finite value $v_0\,\mathrm{Sp}_L/(r_b C_{L,\max})$, so the density is smooth through zero spin. As $|\boldsymbol{\ell}| \to C_{L,\max}$ the Jacobian grows without bound: the whole tail of large spins is squeezed into a thin ring below the plateau. The code gives zero weight to $|\boldsymbol{\ell}| \geq 0.999\,C_{L,\max}$, which removes only spins with $\mathrm{Sp}_0 > -\mathrm{Sp}_L \ln 0.001$, far in the tail of the prior. With Eq. (22), every statement about $\boldsymbol{\ell}$ is a statement about the spin under the prior of Assumption A13.

B.2 Adaptive Proposal for the Nuisance Parameters

When the first stage has $N_{\mathrm{eff}} < N_1/2$ (Step 5), the data constrain $\boldsymbol{\phi}$ enough that the prior wastes samples. Each of the eight uniform components is then mapped to the real line by the logit of its position in its range, $z = \ln[s/(1-s)]$ with $s = (\phi - \phi_{\min})/(\phi_{\max} - \phi_{\min})$. A Gaussian with the weighted mean and covariance of the first-stage values of z is fitted, its covariance is multiplied by the square of the widening factor (Section 5.3) so that its tails cover the posterior, and the N_2 second-stage vectors are drawn from it and mapped back; $\omega_{\parallel}$ is still drawn from its prior. The resulting density $q_2(\boldsymbol{\phi})$ is evaluated in the z coordinates, where the uniform prior becomes the product of logistic densities $\prod s(1-s)$. The Jacobian of the logit map is common to p and q_2 and cancels in their ratio.

B.3 Deterministic-Mixture Weights

The final weight of Eq. (14) treats all N samples of both stages as draws from the mixture of the two proposals, the deterministic-mixture weights of Veach and Guibas (1995) and Owen and Zhou (2000). Compared with a weight that uses the prior alone, each weight is multiplied by

$$\frac{p(\boldsymbol{\phi}_i)}{(N_1/N)\,p(\boldsymbol{\phi}_i) + (N_2/N)\,q_2(\boldsymbol{\phi}_i)} \;\leq\; \frac{N}{N_1}, \tag{23}$$

so no weight can explode where the adapted Gaussian is thin, and samples from the prior keep covering the regions the adapted Gaussian misses; if both stages draw from the prior, the factor is one. The mean of the final weights, $\hat{Z} = N^{-1}\sum_i \tilde{w}_i$, estimates the evidence $p(\mathbf{d})$ of the model and its priors up to a common constant; its ratio between two priors is the Bayes factor used to compare spin priors in Section 9.

B.4 Knuckling in the Observation Window

The knuckling force of Eq. (8) already acts between launch and t_2. A non-spinning ball at $30\,\mathrm{m\,s^{-1}}$ is displaced by it by $0.0086\,\mathrm{m}$ per perpendicular axis at t_2, more than the rounding error $\sigma = 2.9\,\mathrm{mm}$. The likelihood of Eq. (13) therefore compares the data with the knuckle-free positions $\mathbf{d}(\mathbf{u}, \boldsymbol{\phi})$ and adds the covariance of this displacement, $\boldsymbol{\Sigma}_K$. In each of the directions $\hat{\mathbf{e}}_1$ and $\hat{\mathbf{e}}_2$ the displacements at t_1 and t_2 have standard deviations $c_1 a_K$ and $c_2 a_K$ and correlation r, where $a_K = a_{30}\,(v_0/30\,\mathrm{m\,s^{-1}})^2\,e^{-\mathrm{Sp}_0/\mathrm{Sp}_K}$ is the knuckling amplitude at launch. Simulating many non-spinning kicks at several speeds gives $c_1 = 0.001114\,\mathrm{s^2}$, $c_2 = 0.004011\,\mathrm{s^2}$ and $r = 0.95$; the displacement along the flight is negligible. In the Gauss–Newton fit (Nocedal and Wright, 2006), a_K is evaluated at the start of each iteration and held fixed within it. The flight to the goal draws a fresh knuckling realisation, so a push the data may have seen in the window is not carried on to the goal; this neglected correlation matters only for kicks with little spin (Section 7.6).

B.5 Further Test Results

The mismatch set of Section 7.1 draws $C_{D,\mathrm{sup}}$ from 0.20–0.26 (prior 0.12–0.20), $C_{L,\max}$ from 0.32–0.42 (prior 0.28–0.36), Sp_L from 0.07–0.25 (prior 0.10–0.20) and ρ from $1.0\,\mathrm{kg\,m^{-3}}$ to $1.3\,\mathrm{kg\,m^{-3}}$ (prior 1.05–1.25); it uses a spin scale of $35\,\mathrm{rad\,s^{-1}}$ instead of 25, a random wind with standard

deviation $1.5\,\mathrm{m\,s^{-1}}$ in each horizontal component, knuckling 1.5 times stronger and timing errors 2.5 times larger, and it rounds the noisy observations to 0.01 m.

Table 15 completes Table 9 with the two predictors left out of it. Gravity alone (M1) removes the largest systematic error of the straight line but none of the curl; the finer observation grid of GRID5 does not help, because the output is still a ninth of the goal.

Table 15: In-model accuracy of the predictors not shown in Table 9; columns as there.

Predictor	Median error	90th pct.	No crossing	68 % cov.	95 % cov.	Right ninth	Time
M1 gravity only	0.39 m	0.64 m	0	–	–	68 %	5.9×10^{-4} ms
GRID5 grid lookup	1.1 m	5.4 m	–	71 %	94 %	45 %	0.0019 ms

Report on the Use of AI

(Generated at finalisation by tools/build_ai_report.py.)